

High-Impact Application Atlas for the Spherical- Support Event Calculus

Physical, biomedical, digital, energy, logistics, civic and research systems - including MRI, lithography, hospital life journeys, grids, pipelines, solar, event conversion, warehousing and documentation.

95

application hypotheses

16

bounded pilot programs

38

benchmark metrics

36

corpus + external
evidence anchors

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1. Executive synthesis and high-impact thesis

The corpus is most useful as a design language for direct state and event queries across instruments, organizations and infrastructure.

The high-impact opportunity is not a universal replacement for physical law, imaging, matrices, sensors or existing institutions. It is a change in **where authority lives**: the authoritative object becomes a typed state-and-event substrate, while images, dashboards, grids, documents and control displays become downstream views. That enables direct questions about state, compatibility, expected transitions, lineage and missing events before a system materializes every frame, voxel, scan, row or document.

Universal query-first application architecture

One typed substrate; many physical and digital embodiments

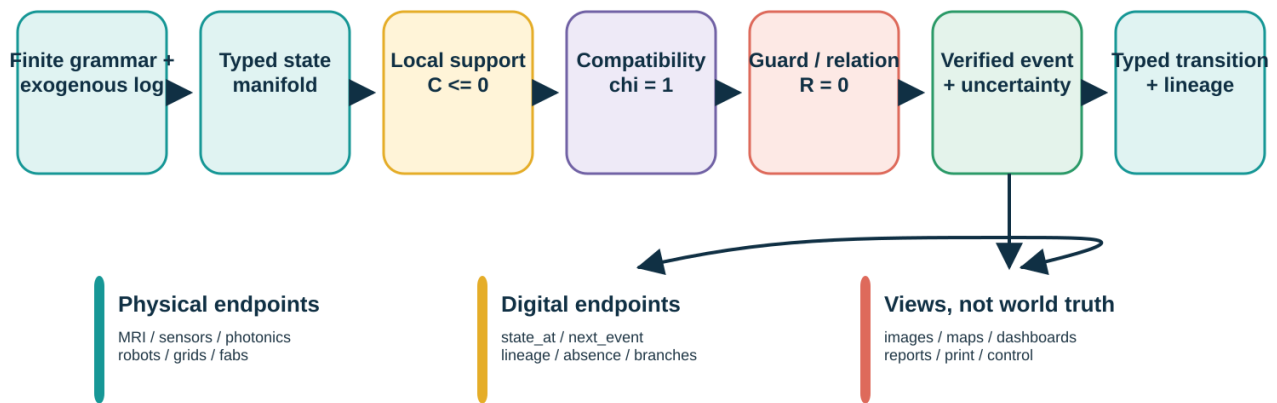
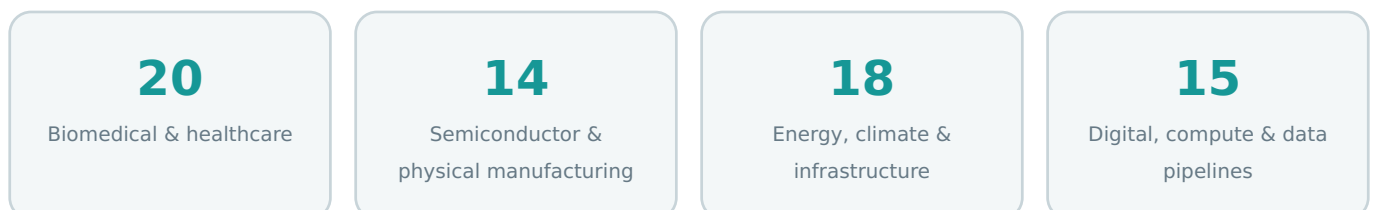


Figure 1. One reusable query/event architecture can sit beneath physical instruments, digital systems and human workflows.



10

Warehousing, logistics &
supply chains

10

Smart cities & municipal
governance

8

Research &
documentation
management

1

shared benchmark
contract

The practical thesis

Use local support to reduce the candidate world; use compatibility to prevent invalid coupling; solve explicit guards for the earliest admitted event; apply a typed transition; preserve lineage; and log external novelty. Only materialize an image, matrix, dashboard or document when the consumer actually needs that view.

Six high-impact mechanisms

- **Direct query instead of universal replay.** For a fixed closed relation family, ask for state at time or next event without replaying every skipped frame.
- **Support-aware computation.** Restrict work to an angular, spatial, causal, semantic or policy support rather than evaluating the whole domain.
- **Compatibility-gated coupling.** Co-location, shared identifiers or shared infrastructure do not automatically authorize interaction.
- **Auditable negative events.** Record when an expected presence failed to occur under valid coverage, instead of confusing absence with missing telemetry.
- **Lineage over snapshots.** Preserve derivation, split, merge, handoff, branch and version history across time.
- **Hybrid physical/digital computation.** Let optics, analog electronics, event sensors or physical transfer functions reject or transform signals before expensive digitization, while a digital sidecar retains calibration, safety, policy and auditability.

Where impact is most plausible first

ID	Flagship application	Practical role	Maturity	Evidence
DIG-08	Auditable absence and negative-event ledger	Addition to any event system.	Implementable design pattern	C0 R6 R7 R24 R26
BIO-01	MRI adaptive acquisition and sequence orchestration	Addition to MRI acquisition and control; not a magnet replacement.	Near-term research pilot	C0 R1 R2 R3 R5
BIO-12	Cradle-to-life longitudinal patient journey substrate	Addition beneath EHRs; not an autonomous medical authority.	Implementable architecture pilot	C0 R6 R7 R8 R28
PHY-02	Computational lithography hotspot query engine	Addition/partial replacement of selected optimization passes.	Research prototype	C0 R12
PHY-06	Fab chamber next-fault digital twin	Addition.	Near-term pilot	C0
ENG-04	Solar PV string and module event twin	Addition.	Near-term pilot	

ID	Flagship application	Practical role	Maturity	Evidence
				C0 R17 R18
ENG-08	Gas, oil, hydrogen and CO2 pipeline leak/rupture event twin	Addition/upgrade to CPM and response coordination.	Near-term research pilot	C0 R19 R20
DIG-04	Event-driven analog-to-digital conversion layer	Addition/partial replacement in sparse sensing.	Near-term hardware research	C0 R31
LOG-01	Warehouse event twin and inventory truth	Addition beneath WMS/WES.	Implementable pilot	C0 R26 R27
CIV-01	Municipal service obligation and 311 event graph	Addition beneath CRM/asset systems.	Implementable pilot	C0 R21 R23 R25
RES-01	Claim-evidence-provenance graph	Addition beneath document systems.	Implementable pilot	C0 R28 R29
DIG-07	Novelty-proportional state storage	Partial replacement of snapshot-heavy history for closed dynamics.	Near-term software pilot	C0 R28

The nontraditional boundary

The same operator vocabulary can link a signal threshold inside an instrument, a patient handoff, a wafer recipe branch, a valve response, a parcel scan, a municipal service deadline and a missing replication study. The systems are not physically identical. What is shared is the information architecture: expectation, support, compatibility, guard, transition, uncertainty and lineage.

2. Evidence boundary and claim discipline

Every application is labeled as corpus-derived, externally established, inferred, pilot-ready or frontier.

The uploaded report explicitly separates a canonical query/event core from speculative physical and biological literalizations. This atlas preserves that asymmetry. The corpus supplies the operator language. External primary or official sources establish existing component classes and standards. The application sections then make cross-domain inferences and define pilots; they do not convert those inferences into measured performance.

Evidence and claim ladder

Keep invention ambitious while making status explicit

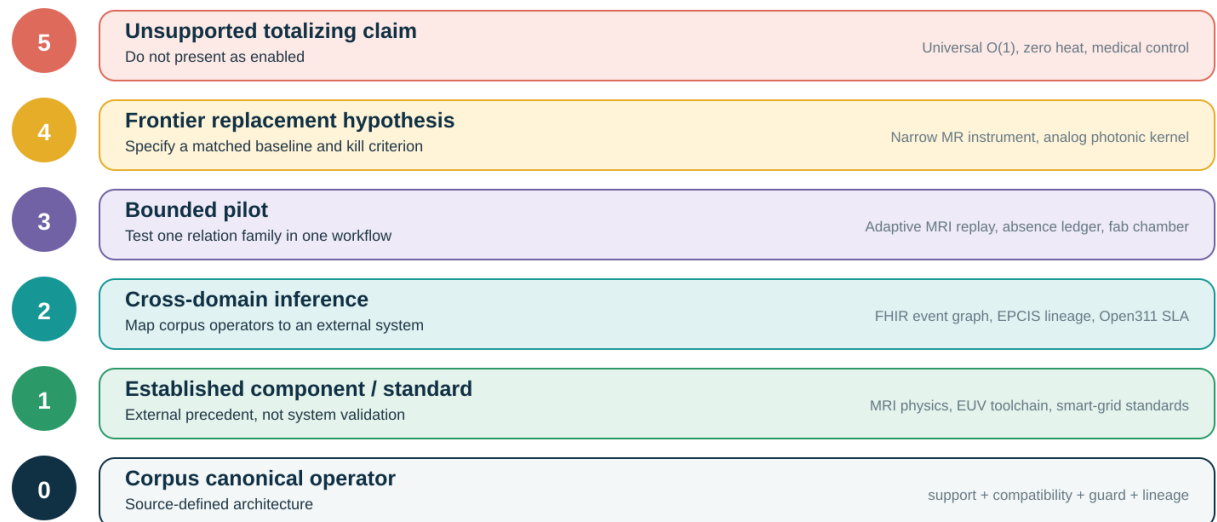


Figure 2. The evidence ladder distinguishes source operators, established components, cross-domain inference, pilots, frontier replacements and rejected totalizing claims.

Source-derived

Query-first authority; product state; relation/support/compatibility conjunction; event transitions; double vacuum as absence of coupling; lineage; novelty log; optional projections; bounded throughput and failure criteria. **co**

External anchors

MRI physics and safety, FHIR, WHO health-AI governance, NIMH RDoC, ASML lithography, NIST smart grid/cities, NREL grid/PV, PHMSA pipelines, GS1 traceability, W3C provenance, event sensors/ADCs and photonic computing. These support components or standards, not the atlas system as a whole.

Cross-domain inference

Examples include adaptive MR acquisition, a cradle-to-life event graph, local lithography hotspot queries, auditable municipal service absence and research claim/evidence graphs. Each needs a domain model, baseline and governance layer.

Rejected shortcuts

No universal $O(1)$; no zero memory, heat, power or latency; no physical shielding from topology; no automatic medical control; no claim that event calculus makes EUV, MRI magnets, PDE solvers or dense linear algebra obsolete.

5. The query-first core

The central inversion is a change in the question asked by the system. A conventional scene pipeline asks what geometry is stored, what must update this frame, and how a sensor produces an image. The equation-world asks which finite relations generate possible states, what state holds at a requested time, which event occurs next, and which sectors are allowed to couple. That is a shift from picture production to direct state semantics.

[S1 pp. 9-10]

Projection-first question

- Which surfaces and samples exist now?
- How are they projected this frame?
- Which pixels or rays are evaluated?
- How is temporal state replayed?

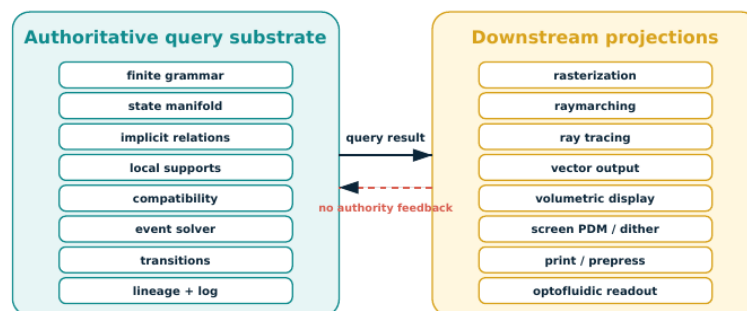
Query-first question

- Which relations define the admissible state?
- What is $q_e(t)$ at requested time t ?
- What is the earliest compatible guard crossing?
- Which identity and invariant survive the transition?

This does not ban graphics. It relocates graphics. A projection may be raster, ray-based, vector, volumetric, print, sensor, or photonic. None of those output forms is allowed to overwrite the authority of the state manifold.

Computer graphics: core versus projection

The corpus overlaps with graphics without reducing the world to drawing.



Do not mistake the projector for the state space.

Figure 5. Query substrate and projection systems are connected, but authority does not flow backward from pixels to ontology.

S1 pp. 9-10; S7 pp. 1-10.

Source report page 13: The query-first core places raster, ray, vector and photonic outputs downstream of the authoritative state substrate.

Report status

This document is an application and experiment atlas. It contains no clinical validation, wafer results, utility deployment results, pipeline certification or municipal policy approval. The full application set should be read as a program of bounded hypotheses with explicit exit criteria.

3. Reusable query-first architecture

A common execution contract can be specialized without pretending that all domains share one physical equation.

State is more than position

A state can carry position, continuous time, phase or mode, sheet or trust context, lineage address, branch/version and a narrow routing bit. In a hospital, sheet may represent consent or care context; in photonics it may be mode or polarization; in a municipal system it may be legal purpose or agency authority. The semantics must be declared per domain.

$$Q = P^n \times R_{\text{time}} \times S^1_{\text{phase}} \times Z_2_{\text{sheet}} \times A_{\text{address}} \times B_{\text{branch}}$$

Normalized product state from the source report; domain implementations may omit or rename coordinates.

An event is a three-gate conjunction

$$t^* = \min \{ t \geq t_0 : R_j(q(t)) = 0 \text{ AND } C_{\alpha}(q(t), t) \leq 0 \text{ AND } \chi(q, j, t, \text{context}) = 1 \}$$

Earliest relation/guard root admitted by local support and compatibility.

The event record must include more than a timestamp. It should preserve the relation and transition identifiers, predicate version, support admission, compatibility reason codes, guard evidence, uncertainty, before/after hashes, lineage, branch creation and invariant checks.

THE EVENT IS A THREE-GATE CONJUNCTION

7. Relation, support, compatibility, event, transition

A pure geometric intersection is insufficient. The source formal core requires three conditions: the relation or guard reaches its event surface; the state lies within a declared local support; and the compatibility predicate admits coupling. Only then is the transition invoked. [S1 pp. 5-7; S6 pp. 3, 5, 7]

RELATION OR GUARD SURFACE

$$R_j(q; \theta_j) = 0$$

LOCAL SPHERICAL, CONIC, SHELL, CAUSAL, OR POLICY SUPPORT

$$C_{\alpha}(q, t) \leq 0$$

COMPATIBILITY PREDICATE

$$\chi(e, j, t, \text{context}) \in \{0, 1\}$$

EARLIEST ADMITTED EVENT

$$t^* = \min \{ t \geq t_0 : R_j(q_e(t)) = 0 \text{ AND } C_{\alpha}(q_e(t), t) \leq 0 \text{ AND } \chi(e, j, t) = 1 \}$$

TYPED TRANSITION

$$q_e(t^+) = T_j(q_e(t^-), \text{context})$$

Why SDF zero matters

Earlier passages sometimes imagine "going past zero" in a continuous field as if that alone escaped rasterization. The mature synthesis corrects this: the zero set is not a magical continuation surface. It is the explicit locus where event semantics are invoked. The engine solves for a root or certified interval and then applies a transition. If it merely samples the field again and again, hidden marching has returned.

[S1 p. 7; S2 pp. 16-19; S3 pp. 30-33]

Topological Cross-Domain Corpus

16

Source report page 16: The source report formalizes the event as relation crossing plus support plus compatibility.

Execution model

1. Evaluate a closed state expression or retrieve a model state at the requested time.
2. Use local support to reduce candidates.
3. Evaluate physical, semantic, policy, consent and identity compatibility.
4. Solve the relation for a root or a certified interval; do not hide a frame loop inside repeated sampling.
5. Apply a typed transition and verify invariants.

6. Append irreducible novelty and provenance to the external log.
7. Generate downstream views for people, machines or records.

Memory model

$M_{\text{required}} \sim \text{seed} + \text{grammar} + \text{external novelty} + \text{branch/lineage history}$

Bounded novelty-proportional memory hypothesis, not a zero-memory theorem.

Predictable closed dynamics may be regenerated. External interventions, sensor observations, policy changes, human decisions and branch history remain stored. The architecture succeeds only when regeneration and event solving cost less than the snapshots or materialized structures it avoids.

11. Memory, lineage, branches, and event sourcing

The corpus proposes that closed dynamics need not be stored frame by frame. A seed and finite grammar regenerate predictable structure; an exogenous log stores only irreducible novelty; branch history and lineage store the information needed to reconstruct splits, merges, and interventions. The mature report treats this as a bounded hypothesis, not a zero-memory theorem. [S1 pp. 7-8; S5 pp. 10-11]

BOUNDED NOVELTY-PROPORTIONAL MEMORY HYPOTHESIS

$M_{\text{required}} \sim \text{seed} + \text{grammar} + \text{exogenous_log} + \text{necessary_branch_history}$

What can be regenerated

- Closed trajectories defined by fixed expressions.
- Implicit geometry defined by finite coefficients.
- Deterministic transitions whose inputs and branch context are known.
- Derived images, meshes, and reports as materialized views.

What still accumulates

- External interventions and sensor surprises.
- Ambiguous or non-deterministic branch choices.
- Predicate and schema revisions.
- Identity split/merge reconciliation.
- Numerical certificates, uncertainty, and calibration state.

This is an event-sourced design pattern: state may be recomputed from compact generative structure plus an authoritative log. The advantage exists only when replay, solving, and branch resolution remain cheaper than storing or materializing the avoided structure. [S1 p. 8]

Lineage rules

```
lineage_address = H(seed, grammar_path, ordered_parent_events)

split(parent, labels):
    create child branches with parent_address and unique branch labels

merge(parents, policy):
    require explicit reconciliation policy
    record all parent addresses and invariant conflicts

reconstruct(address, t):
    load seed + grammar version
    apply admitted exogenous events up to t
    verify invariant and predicate versions
```

Source report page 25: The source report states the bounded memory hypothesis: store irreducible novelty, not every projection.

Type discipline across physical systems

Do not let topology substitute for domain physics

Maxwell still governs electromagnetic fields; Bloch equations and device safety govern MR signal evolution; fluid and surface mechanics govern liquid interfaces; semiconductor process physics governs lithography and deposition; grid equations and protection engineering govern power systems; physiology and clinical evidence govern medicine. The event calculus decides what to query, when a transition is declared and how its evidence is logged.

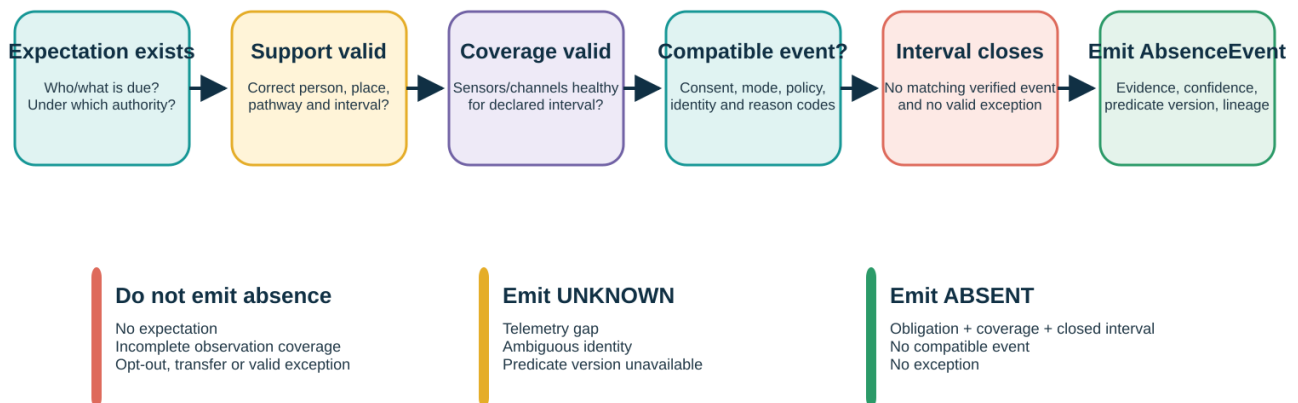
4. Auditable absence: logging expected presence that did not occur

Historical systems often store observed presence and errors, while leaving a required but missing event as a blank or inference.

An absence event is not the absence of a row. It is a positive record created only because a valid expectation existed, the observation support was active, the due interval closed, no compatible event was found and no exception applied.

Auditable absence and the missing-presence problem

Absence is a positive record with obligations and coverage - not a blank database cell



The architecture logs the absence of a required presence without confusing it with absence of observation.

Figure 3. The state machine prevents missing telemetry, opt-out or transfer from being mislabeled as true absence.

AbsenceEvent = ExpectedEvent AND CoverageComplete AND IntervalClosed AND no MatchingCompatibleEvent AND no ValidException

The predicate version, reason codes, authority, confidence and coverage evidence are part of the record.

Why this changes multiple sectors

Domain	Expected presence	Coverage/support contract	Possible negative event	Do not confuse with
Hospital follow-up	Follow-up visit or contact due	Care pathway, consent, contact channels valid	Absent follow-up	

Domain	Expected presence	Coverage/support contract	Possible negative event	Do not confuse with
				Do not emit if patient opted out, transferred care or contact coverage failed.
MRI sequence	Physiology or calibration event required	Sensor attached, channel healthy, sequence window active	Missing required acquisition support	Result is inconclusive, not negative anatomy.
Warehouse	Expected pallet scan at zone exit	Reader coverage and task assignment valid	Unobserved movement or process exception	Distinguish reader failure from item absence.
Delivery	Expected handoff by promise window	Vehicle, parcel and route tracking valid	Failed/late delivery	Recipient unavailability is a reason code, not silent failure.
Pipeline	Expected valve closure after rupture identification	Valve command path and feedback sensor healthy	Mitigation failure	Escalate; do not infer from pressure alone.
Solar	Expected generation under irradiance support	Irradiance, inverter availability and telemetry valid	Underperformance/fault candidate	Night, curtailment and telemetry outage are separate sectors.
Transit	Scheduled trip arrival	AVL coverage and trip not cancelled	Missed trip	Unknown if AVL is down or detour unreported.
Municipal service	Inspection or repair promised	Work order active, access available	Overdue service	Waiver, resident access failure and weather delay are typed reasons.
Research	Pre-registered observation expected	Protocol power and instrument detection support valid	Supported null/non-observation	Instrument failure or underpowered design remains unknown.
Data pipeline	Expected artifact/test/approval	Job and repository available	Blocked release	Differentiate queued, failed, skipped and never required.

Governance requirements

- **Expectation authority.** Identify who or what legitimately created the obligation. A system should not manufacture duties from convenience.
- **Coverage contract.** Record sensor health, channel reachability, time synchronization, identity resolution and blind spots.
- **Exception vocabulary.** Opt-out, transfer, recipient unavailable, planned maintenance and scope exclusion require reason codes.
- **Unknown is a first-class outcome.** Incomplete coverage, ambiguous identity or unavailable predicate versions must stay unknown.
- **Appeal and correction.** Human-facing absence events should support correction, contextualization and provenance-preserving amendment.

Mental-health caution

Silence, non-response, missing diary entries or absence from a platform must never be converted directly into diagnosis, relapse, risk or non-compliance. Without valid coverage, consent, context and clinical interpretation, the only defensible outcome is unknown.

5. Replacement, addition and frontier strategy

The fastest path is to add the event substrate beneath existing systems, then replace only the stages whose materialization is demonstrably redundant.

Replacement versus addition spectrum

The practical program is selective: retain physical law, replace redundant materialization, add auditable event semantics

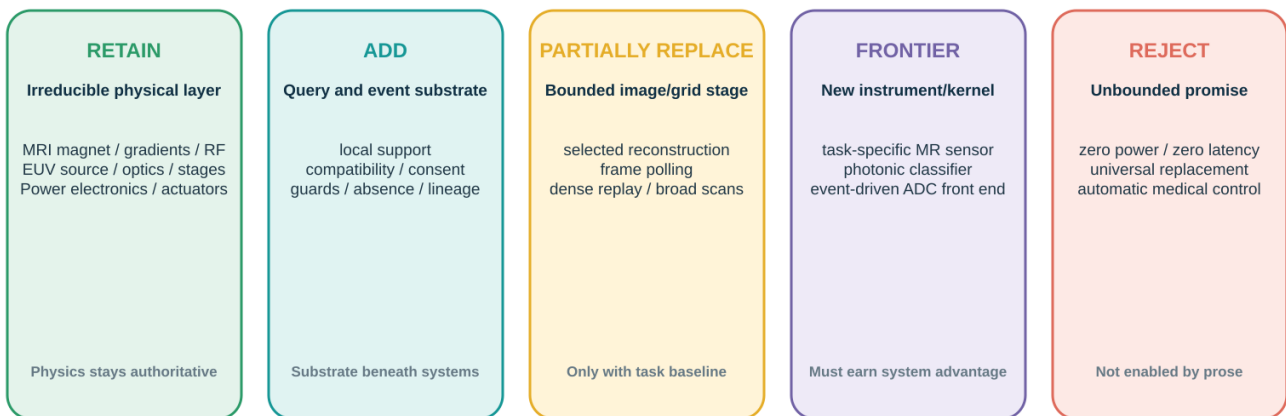


Figure 4. A selective replacement ladder keeps irreducible physical layers and rejects unbounded promises.

Five dispositions

Retain the physical layer

Magnets, RF coils, EUV sources, mirrors, stages, sensors, power electronics, valves, actuators, chemistry and safety systems remain governed by their own physical models.

Add a query/event substrate

Introduce support, compatibility, guards, uncertainty, lineage, negative events and direct state queries beneath current interfaces.

Partially replace bounded stages

Bypass full image formation, frame replay, broad scans, snapshot history or document-centric coordination only for validated narrow tasks.

Build frontier instruments

Explore event ADCs, task-specific MR instruments, photonic transfer kernels and optofluidic prefilters with full calibration and energy accounting.

Reject totalizing replacement

Do not claim that one geometric metaphor replaces every renderer, matrix solver, hospital, chip fab or energy system.

The replacement test

$$\text{Cost}(\text{support} + \text{compatibility} + \text{event solve} + \text{lineage}) < \text{Cost}(\text{materialization avoided})$$

The comparison must be made at equal task correctness, uncertainty, safety and coverage.

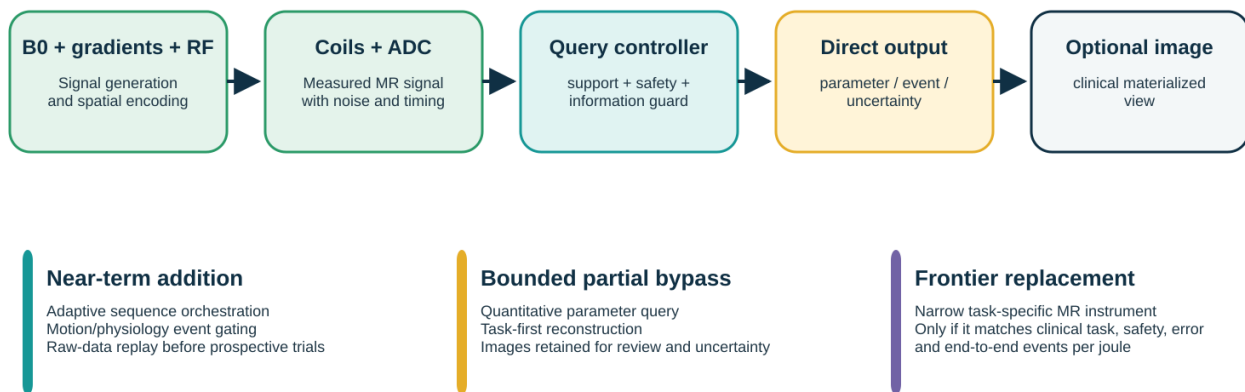
That test is the common boundary for MRI reconstruction, lithography optimization, graphics, state history, warehouse polling and municipal dashboards. An elegant query language that merely moves the same cost elsewhere is not a system improvement.

6. Biomedical, MRI, hospital and mental-health applications

The safest high-impact path is to improve acquisition, continuity, consent, care-gap detection and research provenance - not to turn a topology metaphor into a medical treatment.

MRI: additions, partial bypasses and frontier instruments

The event calculus changes acquisition and state semantics; it does not abolish MR physics or safety



Never infer negative anatomy from missing acquisition support.

Figure 5. MRI can gain a query/event control layer and bounded direct outputs while retaining MR physics and safety.

MRI: from image-first to task-and-event-first

Standard MRI uses a strong static magnetic field, gradients, RF excitation, receiver coils, digitization and reconstruction. This atlas does not remove those layers. It proposes three escalating interventions:

1. **Adaptive acquisition addition.** Use anatomical support, physiology, motion, implant/safety compatibility and uncertainty guards to decide which encoding block is acquired next.
2. **Bounded reconstruction bypass.** For a narrow quantitative task, estimate a parameter or event directly from physics-constrained measurements, while retaining an image and conventional pipeline as a reference.
3. **Frontier task-specific MR instrument.** Build a smaller or simpler instrument only for a declared question and prove task equivalence, safety, uncertainty and events per joule. This is not a general MRI replacement.

MRI safety boundary

Adaptive behavior must remain inside RF, gradient, implant, device-labeling and patient safety constraints. Missing calibration or physiology data is not negative anatomy; it makes the query unsupported or unknown. **R1 R2 R5**

High-impact MRI and imaging applications

BIO-01 MRI adaptive acquisition and sequence orchestration

Near-term research pilot

Addition to MRI acquisition and control; not a magnet replacement.

Role. Use local anatomical/physiological support, motion and safety compatibility, and quantitative guard conditions to adapt which measurements are acquired next.

Guard. Acquire the next encoding block when predicted information gain or quantitative uncertainty crosses a declared threshold and safety constraints remain satisfied.

Auditable absence. Expected calibration, physiological gating or coil data fail to arrive within the sequence window despite verified sensor coverage.

Metrics. Diagnostic task performance; scan time; acquired samples; SAR and gradient limits; motion artifacts; events/j; false-adaptation rate.

Evidence: **C0 R1 R2 R3 R5**

BIO-02 MRI physics-constrained direct parameter query

Research prototype

Partial replacement of selected reconstruction stages for bounded tasks.

Role. Represent Bloch-model-compatible signal families as relations and query selected parameters or change events directly, optionally using images as a view.

Guard. Parameter posterior or certified interval enters a clinically meaningful range under sufficient acquisition support.

Auditable absence. A parameter cannot be certified because the required excitation/encoding support was never observed.

Metrics. Parameter bias/variance; calibration; scan time; task AUC; reconstruction energy; uncertainty coverage.

Evidence: **C0 R3 R4**

BIO-03 MRI narrow diagnostic instrument as an alternative to full imaging

Frontier concept

Alternative narrow instrument; full MRI replacement unsupported.

Role. Design a task-specific MR instrument that answers a bounded question such as relaxation change, flow threshold or lesion-presence probability without requiring a full diagnostic image.

Guard. A task-specific signal relation crosses a validated threshold with confidence and safety conditions satisfied.

Auditable absence. The expected signal support is absent, but only after coil, sequence and coverage integrity are verified.

Metrics. Sensitivity/specificity; acquisition time; energy; footprint; cost; inconclusive rate; referral-to-full-MRI rate.

Evidence: **C0 R1 R4 R5**

BIO-04 MRI motion, respiratory and cardiac event gating

Near-term pilot

Addition.

Role. Treat motion/physiology as continuous state and solve for compatible acquisition windows or correction events rather than replaying uniform frames.

Guard. Predicted motion enters an admissible window and a sequence block is admitted; otherwise hold or redirect.

Auditable absence. A required quiet-window event never occurs within the allowed interval.

Metrics. Rejected acquisitions; reacquisition rate; image/parameter quality; latency; patient time.

Evidence: **C0 R1 R3**

BIO-05

MRI safety and implant compatibility engine

Implementable integration

Addition to safety workflow.

Role. Model implant identity, field strength, sequence limits and patient state as a compatibility predicate with provenance and explicit reason codes.

Guard. Scan becomes admissible only when all required device attributes and MR-conditional constraints are present and current.

Auditable absence. Missing implant identity or condition is logged as a blocking absence, not silently treated as safe.

Metrics. Screening completeness; prevented unsafe scans; delay time; override rate; auditability.

Evidence: C0 R5 R6

BIO-08

Microscopy and cell-culture event microscope

Near-term research pilot

Addition and, for narrow assays, partial replacement of continuous imaging.

Role. Use local optical supports, event sensors or analog prefilters to emit only lineage-aware cell events such as division, migration, morphology change or fluorescence threshold crossing.

Guard. A tracked cell lineage crosses a morphology, intensity or interaction guard with sufficient optical support.

Auditable absence. Expected cell or marker observation absent only when field coverage and sensor health are known.

Metrics. Events/J; bytes/experiment; missed/false events; lineage accuracy; phototoxic exposure.

Evidence: C0 R30

Cradle-to-life hospital journey

Cradle-to-life health journey as a consented event and lineage graph

A person is not a snapshot; care episodes, measurements, decisions and absences remain linked without becoming the person

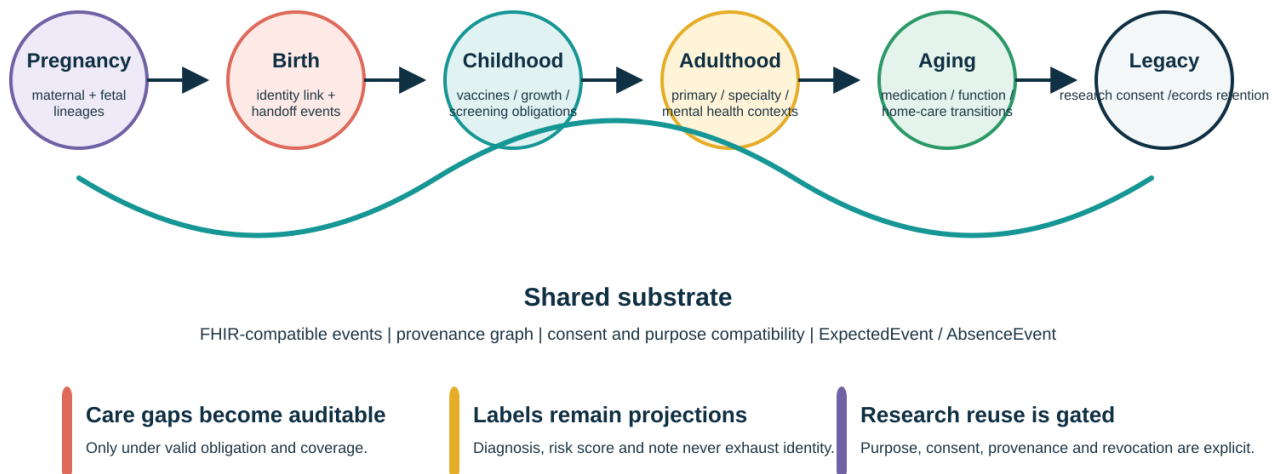


Figure 6. A life-course event graph links care without reducing the person to a single episode, label or snapshot.

The patient journey can be represented as a lineage graph across pregnancy, birth, neonatal care, childhood, adult care, mental health, aging, research consent and record retention. Existing FHIR resources and subscriptions remain exchange mechanisms; the atlas adds expectation, support, compatibility, provenance and branch semantics beneath them. R6 R7 R28

What becomes queryable

- state of a care plan at a requested time;
- next required handoff or follow-up;
- which team is authorized to couple to which context;
- whether a care gap is truly absent or merely unobserved;
- how a diagnosis, medication or sample was derived and changed.

What remains human and institutional

- clinical judgment and shared decision-making;
- consent, capacity and legal authority;
- diagnostic standards and treatment evidence;
- appeal, correction and compassionate context;
- physical care, staffing and resource constraints.

BIO-09 ICU and ward event-native patient monitoring

Near-term pilot

Addition to monitoring and clinical decision support.

Role. Maintain a continuous patient state, local physiological supports, compatibility with interventions and an auditable event record for threshold crossings and trends.

Guard. A validated physiological relation crosses a severity guard and matches the patient context, sensor quality and treatment phase.

Auditable absence. Expected vital, medication response or reassessment is absent despite confirmed observation coverage.

Metrics. Alarm precision/recall; time to recognition; clinician workload; missed-data rate; intervention outcomes.

Evidence: C0 R6 R7 R8 R10

BIO-11 Medication, infusion and closed-loop therapy guard engine

Near-term pilot

Addition.

Role. Join medication identity, dose lineage, patient state, timing and device status through compatibility and event guards; keep clinicians in the loop.

Guard. Dose or infusion transition is admitted only when patient, order, pump, drug, route and timing are compatible.

Auditable absence. Expected reassessment, pump acknowledgment or lab result is missing after its due interval.

Metrics. Medication errors; delayed doses; false blocks; response-monitoring completeness; audit time.

Evidence: C0 R6 R7 R8

BIO-12

Cradle-to-life longitudinal patient journey substrate

Implementable architecture pilot

Addition beneath EHRs; not an autonomous medical authority.

Role. Use a stable person lineage with prenatal, birth, neonatal, childhood, adult and elder branches; represent encounters and documents as views over a consent- and provenance-aware event graph.

Guard. Care transitions, referrals, procedures, consent changes and outcomes are typed events tied to the person lineage.

Auditable absence. Expected follow-up, screening, handoff, result or support service is absent only under an explicit obligation and coverage rule.

Metrics. Record-linkage errors; care-gap closure; duplicate testing; handoff failures; patient access; consent compliance.

Evidence: C0 R6 R7 R8 R28

BIO-13

Maternal, fetal, birth and neonatal journey coordination

Near-term information-system pilot

Addition.

Role. Model mother, fetus/newborn, samples, devices and care teams as related but distinct lineages, with consent and event transitions at birth.

Guard. Birth, transfer, screening, feeding, medication and follow-up transitions preserve parent/child lineage and provenance.

Auditable absence. Expected screening or post-discharge follow-up is absent with verified obligation and outreach coverage.

Metrics. Missed screens; duplicate identity; handoff delay; maternal/newborn linkage accuracy.

Evidence: C0 R6 R8

BIO-14

Hospital flow, beds, theatres and discharge event twin

Near-term pilot

Addition/partial replacement of dashboard-centric coordination.

Role. Represent patient flow, rooms, staff, equipment and constraints as a compatibility-gated event system with next-event and counterfactual queries.

Guard. A transfer or discharge becomes admissible when clinical, staffing, bed, transport and documentation conditions are satisfied.

Auditable absence. Required consult, result, transport or discharge action has not occurred by the due interval.

Metrics. Length of stay; boarding time; cancelled procedures; idle beds; false-ready rate; staff burden.

Evidence: C0 R6 R7 R10

BIO-16

Clinical trial and observational study event graph

Implementable pilot

Addition/partial replacement of document-centric trial operations.

Role. Create a versioned event and provenance graph connecting protocol, consent, intervention, measurement, deviation, analysis and publication.

Guard. Every trial action creates a typed event with protocol version, authority, subject, time and before/after state.

Auditable absence. Expected visit, sample, outcome or adverse-event assessment is explicitly expired, waived or observed.

Metrics. Protocol deviations; reconciliation time; missing data; audit readiness; reproducibility.

Evidence: C0 R6 R28 R29

BIO-17**Biomedical sample and biobank lineage**

Implementable integration

Addition.

Role. Use lineage addresses and split/merge rules for specimen, aliquot, derivative, assay and storage events.

Guard. A sample transition is admitted only when identity, consent, chain of custody, storage and assay compatibility are satisfied.

Auditable absence. Expected receipt, temperature record or assay result is missing after verified custody.

Metrics. Misidentification; unusable samples; audit time; consent violations; freezer excursions.

Evidence: C0 R26 R27 R28

BIO-20**Public-health event and service-absence observatory**

Near-term data pilot

Addition.

Role. Use event notifications, local geographic supports, lineage and explicit expected-report contracts to distinguish true absence, delayed data and sensor failure.

Guard. Case, lab, wastewater, capacity or intervention events cross a public-health guard.

Auditable absence. Expected report absent only after source coverage, reporting obligation and deadline are verified.

Metrics. Reporting delay; missingness classification; false alarms; privacy; action time.

Evidence: C0 R6 R7 R8 R24

Mental-health care and research

The strongest application is a patient-controlled longitudinal substrate for goals, context, consent, outcomes, interventions, research measures and care access. It can support dimensional research and multi-timescale trajectories, but it must not convert platform behavior into diagnosis or automate coercive treatment. WHO governance principles and NIMH RDoC provide useful external boundaries. R8

R9 R10 R11

Prohibited biomedical shortcuts

No direct biological control from phase tags; no autonomous CRISPR or therapy; no medical conclusion from absent telemetry; no replacement of a clinician with an event predicate; no claim that protein folding, cancer or mental illness becomes exactly solvable because a state is represented continuously.

BIO-15

Mental-health longitudinal care and research substrate

Research and governance pilot

Addition to records and research; not automated diagnosis or therapy.

Role. Maintain patient-authored goals, consent, context, development, environment, observations and interventions as lineage-aware events and dimensional trajectories.

Guard. A care or research transition is admitted under explicit consent, role, time and safety predicates, with human review.

Auditable absence. Expected check-in or follow-up may be logged as absent, but lack of digital activity is never equated with deterioration without corroboration.

Metrics. Continuity; patient control; follow-up completion; bias; clinician burden; data access.

Evidence: C0 R8 R9 R10 R11

BIO-18

Protein dynamics and molecular-event simulation

Research prototype

Hybrid addition; broad replacement unsupported.

Role. For selected relation families, solve binding, contact, conformational and threshold events directly, while retaining conventional simulation for unresolved regions.

Guard. Earliest compatible contact, barrier crossing or conformational transition within a bounded model.

Auditable absence. A predicted event is absent only when the modeled support and force field are valid; otherwise unknown.

Metrics. Event-time error; energy conservation; missed states; compute/J; expression growth.

Evidence: C0

BIO-19

CRISPR and gene-editing in-silico candidate pruning

Research concept

Addition to design; not physical gene control.

Role. Use local genomic/epigenetic support, compatibility predicates and lineage to rank candidate interventions and downstream event risks before laboratory testing.

Guard. Candidate passes sequence, structure, cell-context, off-target and policy guards with uncertainty.

Auditable absence. Missing cell-context evidence is a blocking absence, not proof of safety.

Metrics. Off-target ranking; experimental hit rate; uncertainty calibration; provenance.

Evidence: C0 R8 R9

7. ASML, semiconductor and physical manufacturing applications

The architecture can reduce candidate work, improve event timing and preserve process lineage around equipment whose physical functions remain indispensable.

ASML / lithography: event-native augmentation

Change control, optimization and provenance - not the physical creation of EUV light, mirrors, stages or resist chemistry

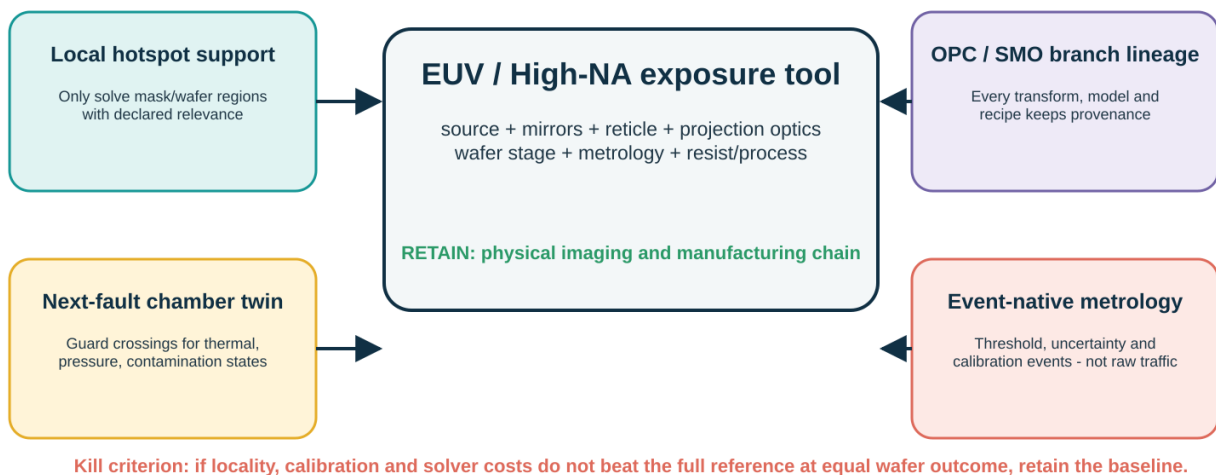


Figure 7. The ASML/EUV opportunity is augmentation around the exposure tool, not a claim that optics and lithography disappear.

ASML and computational lithography

EUV lithography depends on an EUV source, reflective optics, reticle, projection system, stage control, metrology and process chemistry. Computational lithography already uses physics models and optimization. The atlas proposes a query-native layer around these systems: local support for hotspot work, versioned branch lineage for mask and process transforms, guard-based chamber twins and verified metrology events. R12 R13 R14

Plausible additions

- local mask/wafer support to avoid irrelevant solver work;
- event roots for stage, thermal, focus or contamination thresholds;
- provenance for OPC/SMO model, recipe and calibration branches;
- direct pass/fail relation queries for narrow metrology tasks;
- photonic or analog prefilters before high-rate digitization.

What this does not establish

- that legacy nodes replace leading-edge lithography;
- that slower chips are sufficient for all workloads;
- that local support preserves every long-range optical effect;
- that photonics has zero energy or calibration cost;
- that a topology metaphor supplies semiconductor process physics.

From fab to robot cell

PHY-01

ASML/EUV event-native exposure and scanner control

Research/industry pilot

Addition to scanner control; physical exposure system remains.

Role. Add a query-first state/event layer for wafer-stage trajectories, dose, focus, overlay, contamination and subsystem readiness.

Guard. Expose only when stage, source, reticle, wafer, focus, dose and safety states are compatible; latch deviations as events.

Auditable absence. Expected metrology, droplet/source, stage or vacuum readiness event is absent within the control window.

Metrics. Overlay; focus/dose error; wafers/hour; downtime; false aborts; control energy.

Evidence: C0 R12 R13 R14

PHY-02

Computational lithography hotspot query engine

Research prototype

Addition/partial replacement of selected optimization passes.

Role. Represent pattern constraints and process windows as local supports and guards; query hotspot events and compatible corrections instead of materializing every candidate globally.

Guard. A local pattern/process relation exits the manufacturable window under a declared model.

Auditable absence. No hotspot observation is valid only under verified model coverage; unmodeled pattern classes remain unknown.

Metrics. Runtime; memory; hotspot recall; process-window improvement; correction complexity.

Evidence: C0 R12

PHY-03

Reduced-order Maxwell/ rigorous simulation query service

Research prototype

Addition.

Role. Compile bounded parameter families into reduced or surrogate relation models with certified validity regions and fall back to full solvers outside support.

Guard. Parameter query is answered only within certified support and error tolerance.

Auditable absence. A result is absent/inconclusive when the query lies outside the reduced model support.

Metrics. Error bound; solver speed; fallback rate; memory; process-window coverage.

Evidence: C0 R12 R13

PHY-04

Reticle, wafer and process lineage graph

Implementable integration

Addition beneath MES/yield systems.

Role. Use immutable lineage, split/merge and event records across design, mask, exposure, etch, metrology and yield.

Guard. Every custody, recipe, exposure, inspection and disposition creates a versioned event.

Auditable absence. Expected metrology, hold release or lot movement is absent after a declared due interval.

Metrics. Root-cause time; genealogy completeness; scrap containment; auditability.

Evidence: C0 R26 R27 R28

PHY-05

Wafer-stage and robot continuous motion scheduling

Near-term research pilot

Addition.

Role. Use continuous trajectories, local supports and exact/conservative time-of-impact events rather than only fixed-rate checking.

Guard. Earliest contact, clearance or timing-constraint crossing under bounded trajectory uncertainty.

Auditable absence. Tracking absence blocks motion; it never implies free space.

Metrics. Cycle time; missed collision probability; false stops; timing jitter.

Evidence: C0 R33 R34

PHY-06

Fab chamber next-fault digital twin

Near-term pilot

Addition.

Role. Represent pressure, temperature, flow, plasma, chemistry and equipment states continuously; solve next-fault and regime-change events with calibrated uncertainty.

Guard. Earliest compatible threshold or regime transition under a validated physical model.

Auditable absence. Expected sensor or actuator acknowledgment absent with coverage known.

Metrics. Unplanned downtime; false alarms; yield; maintenance hours; model error.

Evidence: C0

PHY-07

Metrology optical prefilter and B.C.E. detector

Frontier lab prototype

Addition; narrow analog front end.

Role. Use calibrated waveguide/optofluidic transfer functions and compatibility gates to detect narrow optical events before full digitization.

Guard. Selected wavelength/polarization/mode response crosses a calibrated guard.

Auditable absence. Expected reference-channel event absent indicates drift or hardware fault, not sample absence.

Metrics. Verified events/J; insertion loss; miss/false rate; calibration interval; throughput.

Evidence: C0

PHY-09

Pick-and-place event-native cell

Near-term pilot

Addition.

Role. Combine continuous motion, local supports, compatibility and lineage for part/nozzle/feeder events.

Guard. Pick or placement is admitted when pose, part identity, nozzle, feeder, board and clearance states match.

Auditable absence. Expected component present but not observed under verified camera/feeder coverage.

Metrics. Cycle time; misspicks; placement error; stoppages; energy/placed part.

Evidence: C0 R26 R33 R34

PHY-11

CNC and tool-wear next-event control

Near-term pilot

Addition.

Role. Represent tool state, workpiece lineage and continuous load/vibration relations; predict wear threshold and collision events.

Guard. Predicted wear, chatter, overload or collision guard crossing.

Auditable absence. Expected cut completion or sensor signature absent under verified coverage.

Metrics. Tool life; scrap; downtime; false alarms; energy/part.

Evidence: C0

PHY-13

Quality inspection as direct relation query

Near-term pilot

Addition/partial replacement for narrow inspections.

Role. Use local support and event-based or analog prefilters for specific geometry, surface or assembly relations; retain images only for exceptions.

Guard. Feature deviation crosses tolerance under sufficient illumination and pose support.

Auditable absence. Expected feature not observed under verified coverage; otherwise unknown.

Metrics. Bytes/part; events/J; miss/false; inspection time; exception-image rate.

Evidence: C0 R30

PHY-14

Photonic relation kernel for fixed industrial queries

Frontier prototype

Addition; domain-specific accelerator.

Role. Map stable linear transforms to calibrated interferometric hardware, while nonlinear logic, uncertainty, lineage and policy remain digital.

Guard. Optical output crosses a calibrated guard and digital sidecar validates context.

Auditable absence. Reference-channel absence or drift blocks result.

Metrics. End-to-end events/J; accuracy; calibration; insertion loss; latency.

Evidence: C0 R32

PCB, pick-and-place and continuous motion

For bounded polynomial trajectories, continuous collision detection and exact or conservative time-of-impact methods provide established precedent for solving contact events rather than hoping a frame sample catches them. The atlas adds compatibility, support and lineage so that a nozzle, component, trace, tool and process recipe are not treated as anonymous coordinates. R33 R34

Manufacturing kill criterion

If candidate pruning misses long-range effects, root solving becomes numerically unstable, calibration dominates, or the electronic/physics reference is cheaper and more accurate, the baseline stays. A narrower successful query is still valuable; it does not have to replace the complete manufacturing stack.

8. Energy grids, pipelines, solar and fossil-fuel reduction

The architecture can improve event detection, dispatch, maintenance, leak response and compute efficiency. Fossil reduction must be counted as measured operational change, not inferred from faster computation.

Energy, grid, pipeline and solar application map

High impact comes from better measurement, dispatch, leak response, maintenance and compute efficiency - not free energy

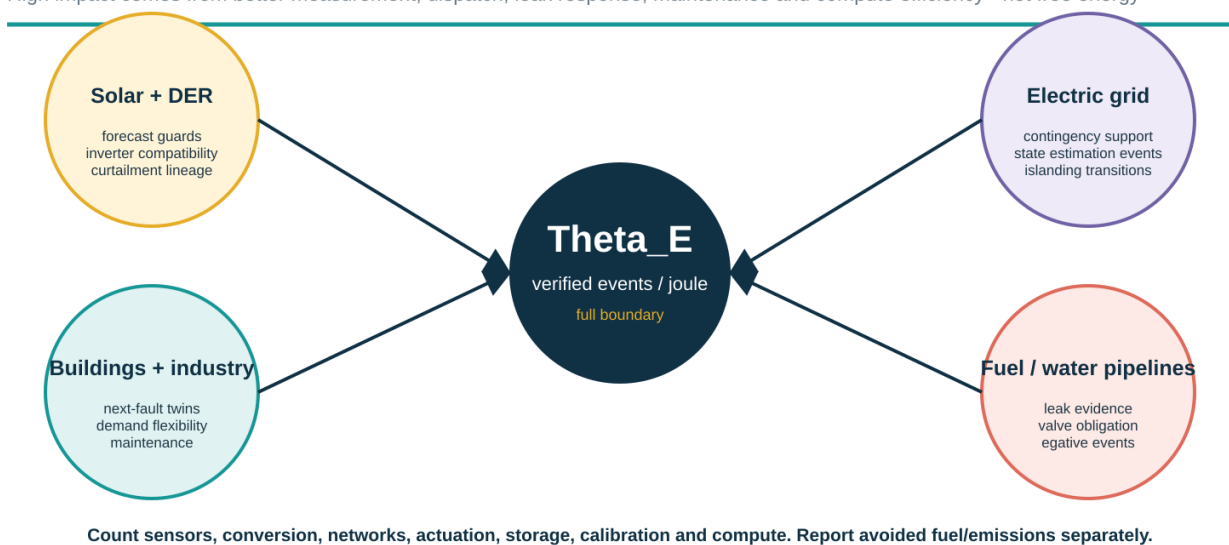


Figure 8. A common event/lineage substrate can connect DERs, grids, facilities and pipelines around a full-system events-per-joule metric.

Grid and distributed-energy applications

NIST smart-grid interoperability and NREL digital-twin work support the need for coordinated state, models and interfaces across heterogeneous grid systems. The atlas adds local contingency supports, compatibility-gated DER interaction, guard-based transitions, counterfactual branches and provenance.

R15 R16 R17

ENG-01**Grid state and contingency query substrate**

Research/utility pilot

Addition beneath EMS/DERMS.

Role. Maintain a versioned grid state and solve local support, compatibility and next-limit events; use conventional power-flow solvers as typed physical services.

Guard. Earliest thermal, voltage, frequency, stability or protection guard crossing under a scenario branch.

Auditable absence. Expected telemetry or breaker state absent is classified separately from a normal zero.

Metrics. Contingency solve time; missed events; false alarms; operator workload; events/J.

Evidence: C0 R15 R16

ENG-02**Distributed energy and microgrid compatibility router**

Near-term pilot

Addition.

Role. Use compatibility predicates for device capability, grid state, market/policy and ownership; route only admitted events.

Guard. Dispatch or islanding transition admitted when electrical, contractual, cyber and timing predicates match.

Auditable absence. Expected device acknowledgment or telemetry absent after a verified communication window.

Metrics. Dispatch success; interoperability failures; curtailment; resilience; cyber incidents.

Evidence: C0 R15 R17

ENG-03**Protection and disturbance event engine**

Near-term research pilot

Addition to analysis and coordination, not unvalidated relay replacement.

Role. Represent disturbance supports, device compatibility, guard crossings and event lineage explicitly; keep certified relays authoritative.

Guard. Fault or instability relation crosses a certified boundary in the protected zone.

Auditable absence. Expected breaker trip or relay acknowledgment absent within the mandated interval.

Metrics. Detection time; selectivity; false trips; coordination; event reconstruction.

Evidence: C0 R15

ENG-04**Solar PV string and module event twin**

Near-term pilot

Addition.

Role. Use local array/string support, expected-yield relations, inverter compatibility and lineage to detect shading, soiling, mismatch, connector and degradation events.

Guard. Measured/predicted energy balance exits a calibrated support or a component relation crosses a fault guard.

Auditable absence. Expected generation or telemetry absent only after irradiance, availability and sensor coverage are verified.

Metrics. Energy recovered; detection delay; false alarms; O&M cost; events/J; curtailment avoided.

Evidence: C0 R17 R18

ENG-05

Solar inverter and MPPT event-native control

Research pilot

Addition; low-level control remains conventional unless certified.

Role. Add local event rules for mode changes, mismatch, voltage support and thermal limits; investigate level-crossing conversion for sparse supervisory signals.

Guard. Operating point, thermal, grid-support or protection relation crosses a guard.

Auditable absence. Expected inverter response absent under verified communications and power availability.

Metrics. Wh delivered per control joule; tracking efficiency; response time; thermal cycling.

Evidence: C0 R17 R31

ENG-07

EV charging and fleet energy routing

Near-term pilot

Addition.

Role. Use vehicle/charger/grid compatibility, local feeder support and next-departure/constraint events with lineage-aware energy accounting.

Guard. Charge/discharge transition admitted under connector, battery, departure, feeder and tariff constraints.

Auditable absence. Expected vehicle arrival, charger acknowledgment or departure readiness absent.

Metrics. Peak demand; missed departures; energy cost; battery wear; charger utilization.

Evidence: C0 R15

ENG-17

Renewable curtailment and storage counterfactual planner

Near-term planning pilot

Addition.

Role. Use isolated branches over a shared state to ask which intervention prevents the next constraint while preserving provenance.

Guard. Branch-specific constraint or reliability event under a candidate storage, network or demand-response action.

Auditable absence. Expected asset availability absent in a branch is explicit.

Metrics. Curtailment avoided; reliability; cost; model runtime; branch explainability.

Evidence: C0 R15 R16

Solar throughput and maintenance

A PV array can be modeled as strings, modules, inverters, sensors, weather support and maintenance obligations. High impact comes from earlier fault discrimination, reduced truck rolls, lower curtailment, better inverter coordination and auditable recovery of lost generation. The benchmark must include sensing, communications, control and calibration energy, not only inverter switching. R17 R18

Pipelines, leak response and the absence of mitigation

Leak detection systems and rupture-response rules already emphasize monitoring, confirmation and valve response. The atlas makes the obligation explicit: once a rupture condition is admitted, a valve-closure event is expected within a declared interval. If feedback coverage is healthy and closure does not occur, the system emits a mitigation-failure absence event rather than leaving an ambiguous gap. **R19**

R20

ENG-06

Battery state-of-health and thermal-runaway guard engine

Near-term pilot

Addition.

Role. Use continuous state relations, lineage of cells/modules and next-event guards for degradation, imbalance, venting and thermal risk.

Guard. Predicted temperature, voltage, pressure, impedance or imbalance crosses a certified threshold.

Auditable absence. Expected sensor/contactor response absent; never interpret missing telemetry as safe.

Metrics. Prediction horizon; false/late trips; cycle life; pack availability; compute energy.

Evidence: **C0**

ENG-08

Gas, oil, hydrogen and CO2 pipeline leak/rupture event twin

Near-term research pilot

Addition/upgrade to CPM and response coordination.

Role. Express leak/rupture, negative-pressure-wave and valve-response events with support, compatibility, uncertainty and auditable negative events.

Guard. Hydraulic/mass-balance or wave relation crosses a leak guard and is compatible with operating transients.

Auditable absence. Expected valve closure, pressure recovery or sensor report absent within a verified interval.

Metrics. Leak size/time sensitivity; location error; false alarms; isolated volume; response time.

Evidence: **C0** **R19** **R20**

ENG-09

Water distribution leak and service-loss event system

Near-term municipal pilot

Addition.

Role. Combine hydraulic relations, local geographic support, meter/sensor compatibility and 311 reports; explicitly model service absence.

Guard. Pressure/flow imbalance or service complaint crosses a localized guard.

Auditable absence. Expected pressure, meter heartbeat or repair completion absent under coverage.

Metrics. Water loss; detection time; false alarms; repair time; equity by neighborhood.

Evidence: **C0** **R23** **R24**

ENG-10

Industrial combustion, boilers and process heat next-event optimization

Research pilot

Addition.

Role. Layer a query-first event twin over validated thermodynamic/control models to predict limit crossings, fouling and efficiency transitions.

Guard. Efficiency, emissions, temperature or pressure relation crosses a guard under a candidate operating branch.

Auditable absence. Expected actuator or sensor response absent.

Metrics. Fuel per output; emissions; downtime; false interventions; control energy.

Evidence: **C0**

ENG-14

Wind turbine and rotating-equipment next-fault engine

Near-term pilot

Addition.

Role. Represent phase, rotational state, load and component lineage; solve bearing, imbalance and fatigue guard events.

Guard. Vibration/temperature/load relation crosses a calibrated fault guard.

Auditable absence. Expected phase marker or sensor heartbeat absent.

Metrics. Lead time; false alarms; downtime; maintenance cost; energy availability.

Evidence: C0

ENG-15

District heating, cooling and thermal-network routing

Research pilot

Addition.

Role. Use continuous transport relations, local branch support and compatibility to predict demand/temperature events and route heat.

Guard. Supply/return temperature, pressure or storage state crosses an operational guard.

Auditable absence. Expected demand or telemetry absent under verified coverage.

Metrics. Fuel/electricity; losses; comfort; peak reduction; event accuracy.

Evidence: C0

ENG-18

Environmental sensing with event/absence semantics

Near-term pilot

Addition.

Role. Use local support, sensor provenance, coverage contracts and threshold events; distinguish non-detection from sensor silence.

Guard. Pollutant or ecological indicator crosses a calibrated guard under valid sensor support.

Auditable absence. Non-detection is emitted only when detection capability and sampling coverage are certified.

Metrics. Detection delay; false absence; sensor uptime; spatial coverage; energy.

Evidence: C0 R24

Fossil-fuel reduction pathways

1. **Leak and vent reduction:** earlier evidence fusion, fewer false alarms, faster validated response and explicit missing-mitigation events.
2. **Renewable utilization:** lower curtailment and better storage/DER scheduling under grid compatibility constraints.
3. **Operational efficiency:** next-fault and process-heat queries that reduce wasted fuel, downtime and conservative over-operation.
4. **Compute efficiency:** avoid unnecessary frame, snapshot and full-materialization work in control rooms, digital twins and data centers.

5. **Logistics and maintenance:** route people, spares and inspections to verified events rather than periodic blanket schedules.
6. **Provenance:** link energy and carbon claims to meter, model, dispatch and intervention lineage.

Fossil-reduction boundary

The architecture does not create renewable generation, electrify heat, replace policy, build transmission or change thermodynamic limits. Report avoided fuel and emissions as separately measured outcomes with baselines and uncertainty. **R15** **R16** **R18** **R19**

ENG-11

Fossil-fuel reduction through event-native operations

Portfolio hypothesis

Cross-sector addition; not a substitute for clean-energy policy or physical transition.

Role. Apply event-native sensing, direct queries, provenance and compatibility to reduce avoidable compute, leaks, curtailment, idling and overproduction.

Guard. Avoided-energy or avoided-emission action occurs only when a verified event justifies it.

Auditable absence. Missing data cannot be counted as avoided use or emissions.

Metrics. Net J saved; kgCO₂e avoided; rebound; capital cost; verification confidence.

Evidence: **C0** **R15** **R18** **R19**

ENG-12

Data-center and AI energy workload router

Research prototype

Addition.

Role. Use local workload support, compatibility, event thresholds and hardware-specific energy models to choose electronic, analog or photonic paths.

Guard. Route a task when accuracy, latency, energy, precision and hardware calibration predicates are satisfied.

Auditable absence. Expected accelerator result or calibration signal absent; fall back.

Metrics. J/query; throughput; accuracy; data movement; fallback rate.

Evidence: **C0** **R32**

ENG-13

Building HVAC and indoor-environment event twin

Near-term pilot

Addition.

Role. Use local zone supports, occupancy/privacy compatibility and next-comfort/air-quality/fault events; store exceptions and novelty rather than every derived view.

Guard. Comfort, CO₂, humidity, equipment or demand guard crossing.

Auditable absence. Expected occupancy or sensor signal absent only with coverage/consent known.

Metrics. kWh; comfort; indoor air quality; false occupancy; maintenance.

Evidence: **C0** **R21** **R24**

ENG-16

Carbon and energy provenance ledger

Implementable information-system pilot

Addition.

Role. Use lineage, versioned factors, compatibility and explicit missingness to make every claim reconstructable and branch-aware.

Guard. A carbon/energy claim is emitted only when source data, factor version, scope and ownership predicates are complete.

Auditable absence. Missing activity or factor data remains a declared gap, not a zero.

Metrics. Coverage; audit time; recalculation reproducibility; uncertainty; double-count prevention.

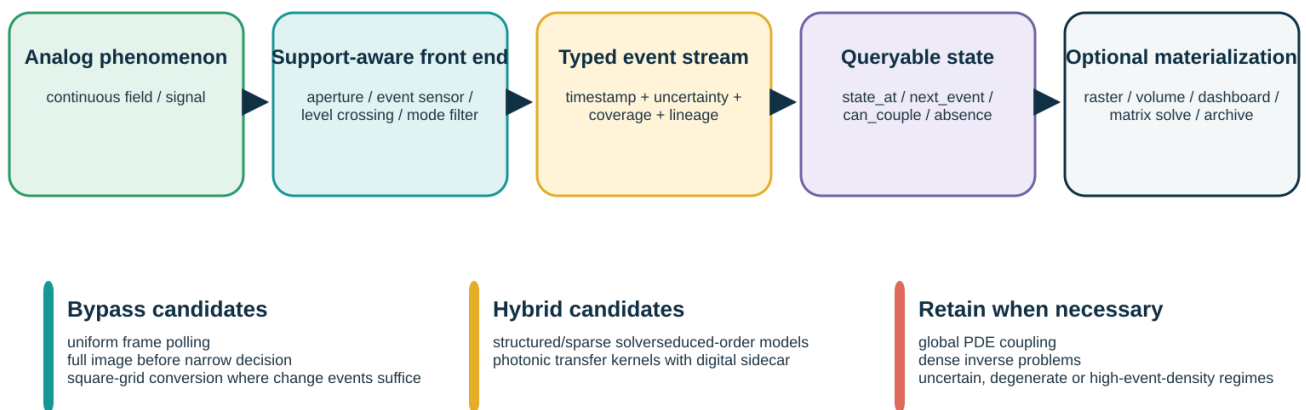
Evidence: **C0** **R28** **R29**

9. Digital pipelines, ADC, graphics and linear-algebra bottlenecks

The design target is less unnecessary materialization: not an anti-matrix, anti-ADC or anti-graphics doctrine.

Digital pipeline: from analog change to direct event semantics

Reduce conversion and materialization where possible; keep dense numerical methods when they remain the right solver



The target is less unnecessary materialization - not the elimination of linear algebra, ADCs or graphics.

Figure 9. Event sensors, level-crossing conversion and query-first state can reduce conversion and materialization before optional images or dense solvers.

Bypassing raymarching and rasterization as world authority

Rasterization, raymarching and ray tracing remain valuable output and transport techniques. The bypass is architectural: collision, state, identity and event semantics should not be reconstructed from pixels when they can be queried directly. A renderer consumes a state query; it does not define the state.

DIG-01 Equation-world simulation kernel

Research prototype

Partial replacement of the authoritative simulation layer in bounded domains.

Role. Implement direct state_at, next_event, support, coupling, routing and lineage queries for a bounded relation family; render only on demand.

Guard. Earliest admitted relation crossing with certified root/interval.

Auditable absence. Expected event absent only with complete relation/support coverage.

Metrics. Query cost; event ordering; memory/novelty; correctness; expression growth.

Evidence: C0

DIG-02 Raymarching/rasterization bypass for state and collision

Research prototype

Partial replacement of state/collision pipeline; rendering remains optional.

Role. Move authority to relations/events and use ray/raster methods only as materialized views; collision and causality use direct queries.

Guard. Visibility, contact or transition query solved from relations rather than image sampling.

Auditable absence. No pixel or sample is never evidence that an entity is absent from state.

Metrics. State query time; collision correctness; render independence; memory; view consistency.

Evidence: C0 R33

DIG-15 AR/VR and simulation foveated query substrate

Research prototype

Addition/partial state-pipeline replacement.

Role. Use gaze/agent-local spherical support for relevance and direct state/event queries; render only admitted views and preserve nonvisual state independently.

Guard. View or interaction event within gaze, reach, causal or semantic support.

Auditable absence. No rendered object is not evidence of world absence.

Metrics. Frame cost; motion-to-photon latency; interaction correctness; energy.

Evidence: C0

Analog-to-digital conversion and event sensing

Event cameras and level-crossing ADCs demonstrate that a sensor can report change events rather than uniformly sampled frames or codes. The atlas extends that idea by attaching support, coverage, compatibility, uncertainty and lineage to each event. Physical or analog preprocessing is useful when the input is sparse and the rejected signal truly need not be digitized. R30 R31

DIG-04

Event-driven analog-to-digital conversion layer

Near-term hardware research

Addition/partial replacement in sparse sensing.

Role. Use level-crossing, change-triggered or feature-triggered conversion for sparse signals, with conventional sampling as fallback and calibration reference.

Guard. Analog signal crosses a calibrated level or derivative guard.

Auditable absence. No event means no threshold crossing only if sensor health, coverage and threshold configuration are known.

Metrics. Samples/signal; energy; reconstruction error; latency; missed crossings.

Evidence: C0 R31

DIG-05

Event-camera perception substrate

Implementable research pilot

Addition/partial replacement for motion-centric tasks.

Role. Use asynchronous change events as input support, then compatibility and lineage to form object/interaction events; images remain optional.

Guard. Local contrast event joins a tracked entity/trajectory and crosses a task guard.

Auditable absence. No events imply no local contrast changes only within sensor support; not object absence.

Metrics. Latency; bandwidth; events/J; tracking accuracy; low-motion failure.

Evidence: C0 R30

DIG-06

Photonic/mixed-signal query accelerator

Frontier prototype

Addition.

Role. Route stable, high-reuse linear transforms to photonic or analog kernels and keep compatibility, uncertainty, nonlinear decisions and lineage digital.

Guard. Hardware path admitted when precision, calibration, batch reuse and I/O energy satisfy a policy.

Auditable absence. Missing calibration/reference event forces digital fallback.

Metrics. End-to-end J/query; throughput; accuracy; I/O share; recalibration.

Evidence: C0 R32

Grids, square matrices and linear-algebra bottlenecks

The corpus language is often read as a promise to bypass gridded or square linear algebra. A bounded interpretation is more useful:

- **Use implicit relations** instead of global meshes when the geometry or state family is compact and queryable.
- **Use local support** to reduce a global operator to a relevant subproblem.
- **Use structured, sparse, low-rank or reduced-order forms** when the physics and error budget permit.

- **Use root/event solvers** for threshold and contact questions rather than stepping every intermediate state.
- **Use analog or photonic transfer functions** for stable narrow transforms, with ADC/DAC, control and calibration included in the energy boundary.
- **Retain dense linear algebra** for globally coupled inverse problems, PDEs, high-event-density regimes and cases where structure is insufficient.

Photonic boundary

Coherent photonic circuits establish a component class for optical matrix-like transforms, not an automatic end-to-end advantage. Sources, modulators, detectors, conversion, control, drift and calibration remain in the benchmark. **R32**

DIG-03 Dense linear-algebra avoidance planner

Research architecture

Addition/partial workload replacement, not elimination of linear algebra.

Role. Use support pruning, typed relations, sparse/structured solvers and task routing; use dense linear algebra only where it is the correct local kernel.

Guard. Choose the smallest valid solver/kernel whose support and precision satisfy the query.

Auditable absence. A solver result is unavailable when support or conditioning is insufficient.

Metrics. FLOPs avoided; data movement; J/query; accuracy; fallback rate.

Evidence: **C0** **R32**

DIG-07 Novelty-proportional state storage

Near-term software pilot

Partial replacement of snapshot-heavy history for closed dynamics.

Role. Store seed/grammar, exogenous events, lineage, branch policy and checkpoints; materialize views on demand.

Guard. An exogenous intervention or irreversible observation is appended as novelty.

Auditable absence. Expected events can expire into absence events; missing coverage remains unknown.

Metrics. Storage growth; rebuild time; auditability; branch cost; view freshness.

Evidence: **C0** **R28**

DIG-08

Auditable absence and negative-event ledger

Implementable design pattern**Addition to any event system.**

Role. Add ExpectedEvent and AbsenceEvent objects requiring obligation, observation support, deadline, coverage and reason codes.

Guard. At deadline, emit absence only if expectation exists, observation support was valid and no compatible event occurred.

Auditable absence. The pattern itself formalizes auditable absence.

Metrics. False absence; unknown-vs-absent classification; overdue time; coverage quality.

Evidence: C0 R6 R7 R24 R26

DIG-09

Data-pipeline state and lineage engine

Implementable integration**Addition/partial replacement of dashboard-centric orchestration.**

Role. Represent datasets, transformations, models, approvals and deployments as typed events with compatibility and provenance.

Guard. Transformation or release admitted only when inputs, schema, tests, authority and policy are compatible.

Auditable absence. Expected artifact, test or approval absent after a due interval.

Metrics. Reproducibility; failure diagnosis; stale-data incidents; audit time.

Evidence: C0 R28 R29

DIG-11

Distributed-system reconstruction and branch debugging

Near-term software pilot**Addition.**

Role. Use explicit event lineage, branch context, compatibility and invariant reports to reconstruct what happened and why.

Guard. State transition creates before/after hashes, predicate version, branch and invariant report.

Auditable absence. Expected acknowledgment absent can be distinguished from delayed, dropped or never-sent.

Metrics. Mean time to resolution; replay fidelity; log volume; invariant violations.

Evidence: C0 R28

DIG-12

AI memory and retrieval with provenance and support

Near-term pilot**Addition.**

Role. Use local semantic support, compatibility, provenance and event-based updates; render summaries only from admitted evidence.

Guard. Evidence enters a response only when relevance, authority, time, permission and provenance predicates pass.

Auditable absence. Expected evidence absent is reported as a gap, not fabricated.

Metrics. Citation accuracy; retrieval precision; stale-source rate; permission violations.

Evidence: C0 R28 R29

DIG-14

Network and telecom event-native monitoring

Near-term pilot**Addition.**

Role. Use local flow supports, compatibility and threshold events; retain raw data around exceptions and provenance.

Guard. Latency, loss, congestion, security or service relation crosses a guard.

Auditable absence. Expected heartbeat or route acknowledgment absent under verified coverage.

Metrics. Data reduction; detection time; false alarms; root-cause time; energy.

Evidence: C0

DIG-13

Model monitoring and deployment event graph

Implementable integration

Addition.

Role. Track data, model, evaluation, approval, deployment and drift as a provenance graph with compatibility and rollback branches.

Guard. Deployment admitted when dataset, model, evaluation, policy and environment are compatible.

Auditable absence. Expected monitoring metric absent triggers a hold, not assumed normality.

Metrics. Rollback time; drift detection; reproducibility; audit completeness.

Evidence: C0 R28 R29

Historical state tracking

Traditional logs often preserve events that happened and periodic snapshots, but not every expected event that failed to happen. The combined model uses a generative state description, external novelty log, lineage and ExpectedEvent/AbsenceEvent records. It can reconstruct a branch while keeping unknown coverage separate from negative evidence.

18. Experiment program and benchmarks

The decisive criterion is not whether the system sounds non-raster. Compatibility plus event solving must grow more slowly than the structure the system avoids materializing. Every benchmark therefore needs an electronic or conventional reference, declared error boundaries, and a kill criterion. [S1 p. 8; S6 pp. 6, 9-10]

Software experiment matrix

Experiment	What it tests	Success evidence	Failure/kill evidence
Horizon skipping	Whether state _{at(t)} cost depends on skipped steps or expression size.	Cost remains tied to expression complexity across larger time horizons.	Expression or branch growth tracks elapsed horizon.
Compatible root solving	Whether support and ch1 prune candidate events enough.	Solver candidate count stays sparse; certified ordering is stable.	Event explosion or broad scan dominates.
Double-vacuum co-location	Whether same-coordinate states remain isolated until admitted.	No cross-sheet event before explicit compatibility.	Implicit coupling leaks across sectors.
Transition routing	Whether parity/sheet changes preserve invariants and lineage.	Deterministic post-state and reconstructable audit.	Ambiguous ordering, lost lineage, or invariant failure.
Grammar depth stress	Whether rule composition stays closed or normalizable.	Expression growth is bounded within declared family.	Closure fails or expressions grow without practical bound.
Split/merge identity	Whether lineage reconstructs nontrivial persistence.	All parents/children and reconciliation decisions are recoverable.	Address collisions or irreversible ancestry loss.
Projection independence	Whether multiple outputs derive from one authoritative query state.	Raster/vector/print outputs disagree only within declared projection error.	Output pipeline silently becomes the authority.

Core software metrics

Metric	Definition	Purpose
Query latency distribution	median, p95, p99 by query class	Separates typical from degeneracy/tail behavior.
Candidate reduction	relations before support / after support / after compatibility	Measures actual pruning value.
Root certification rate	certified events / attempted event queries	Tracks solver reliability.
Miss / false event rate	against exact or high-precision reference	Defines correctness boundary.
Expression growth	nodes/bytes versus grammar depth and branch history	Tests closure and normalization.
Novelty memory	stored log + branch bytes versus materialized baseline	Tests memory hypothesis.
Lineage reconstruction	success, ambiguity, time, collision count	Tests identity semantics.
Projection divergence	difference between views of same query result	Tests downstream-view discipline.

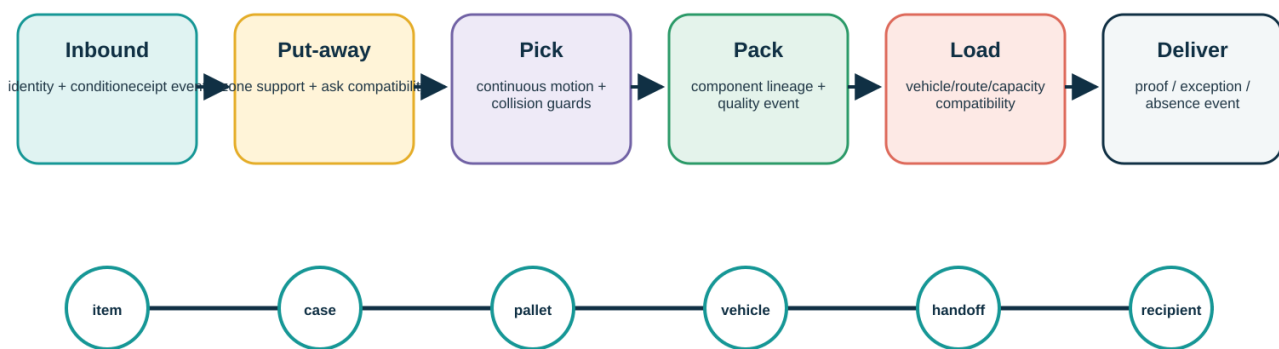
Source report page 38: The source benchmark program tests state-at-time, next-event, support, compatibility, routing and lineage rather than visual fidelity.

10. Warehousing, delivery and supply-chain applications

The high-impact shift is from periodic inventory belief to event lineage with explicit expectations, coverage and custody.

Warehouse and delivery optimization as event lineage

Items, containers, tasks, vehicles and custody changes are explicit relations - not inferred from stale coordinates



Lineage records splits, merges, substitutions and custody - while ExpectedEvent/AbsenceEvent distinguishes late, missing and unobserved.

Figure 10. Item, case, pallet, vehicle, handoff and recipient events preserve lineage across physical movement and process branches.

GS1 EPCIS and traceability standards already provide event and business vocabulary for supply chains. The atlas adds direct state/next-event queries, continuous motion guards, observation coverage, negative events and counterfactual branches. R26 R27

High-impact workflow changes

- **Inventory truth:** treat location as a time-bounded event with reader coverage, not an eternal coordinate.
- **Expected scan semantics:** emit a missing-exit or missing-handoff event only when reader and task coverage were valid.
- **Continuous conflict timing:** solve AMR/forklift time-of-conflict within a local support instead of checking every pair every tick.
- **Custody lineage:** preserve splits, repacks, substitutions, returns, repairs and recalls.
- **Delivery promises:** model proof, exception and recipient availability as typed outcomes rather than one opaque status.
- **Counterfactual routing:** isolate what-if branches from production state while sharing base facts.

LOG-01**Warehouse event twin and inventory truth****Implementable pilot****Addition beneath WMS/WES.**

Role. Use EPCIS-like visibility events, lineage and auditable absence to reconstruct item, tote, pallet, location and task state.

Guard. Receive, move, pick, pack, load, count and exception events update lineage.

Auditable absence. Expected scan or movement absent under verified reader/task coverage.

Metrics. Inventory accuracy; search time; write-offs; reconciliation labor; event completeness.

Evidence: C0 R26 R27

LOG-02**AMR and forklift continuous routing****Near-term pilot****Addition.**

Role. Use continuous trajectories, local supports, compatibility and next-conflict events across vehicles, humans and temporary zones.

Guard. Earliest predicted conflict, blockage or task-window crossing.

Auditable absence. Tracking absence blocks high-speed motion; no return is not free space.

Metrics. Throughput; near misses; stops; travel distance; energy/order.

Evidence: C0 R33 R34

LOG-03**Pick-path and wave planning as event scheduling****Near-term software pilot****Addition.**

Role. Schedule next admissible pick/pack/ship events using local resource support, due-time guards and branch isolation.

Guard. Task admitted when inventory, worker/robot, container, dock and promise constraints match.

Auditable absence. Expected inventory or resource event absent.

Metrics. Orders/hour; lateness; walking/drive distance; replans; fairness.

Evidence: C0

LOG-04**Cold-chain and condition lineage****Implementable pilot****Addition.**

Role. Bind sensor observations, packaging, custody and product lineage; trigger excursion and missing-coverage events.

Guard. Temperature/humidity/time relation exits product support during a custody interval.

Auditable absence. Expected logger data absent under a known custody and power window.

Metrics. Excursion detection; spoilage; false holds; audit time.

Evidence: C0 R24 R26 R27

LOG-05**Last-mile delivery event and promise engine**

Near-term pilot

Addition.

Role. Use continuous vehicle/package state, local traffic support, compatibility and delivery-promise guards with explicit non-delivery reasons.

Guard. Delivery becomes admissible when parcel, recipient/location, vehicle, access and time conditions match.

Auditable absence. Expected arrival, handoff or proof event absent with verified route/coverage.

Metrics. On-time rate; miles/order; failed deliveries; customer contacts; emissions.

Evidence: C0 R26 R27

LOG-06**Port, terminal and yard coordination**

Research/operations pilot

Addition.

Role. Create compatibility-gated events and lineage across berth, yard, crane, customs and onward transport.

Guard. Move admitted when asset, slot, clearance, equipment and safety states match.

Auditable absence. Expected clearance, equipment or handoff event absent.

Metrics. Dwell time; crane productivity; truck queues; misroutes; energy.

Evidence: C0 R26 R27

LOG-07**Food provenance and recall event graph**

Implementable integration

Addition.

Role. Use critical tracking events, split/merge lineage, condition data and compatibility to localize affected product.

Guard. Transformation, aggregation, shipment, receipt and recall events preserve parent/child lineage.

Auditable absence. Expected custody or transformation record absent.

Metrics. Recall scope; trace time; waste; coverage; false inclusion/exclusion.

Evidence: C0 R26 R27

LOG-08**Medical-device and medicine supply chain**

Implementable pilot

Addition.

Role. Combine GS1 traceability with hospital compatibility and patient/procedure context.

Guard. Device/drug use admitted when product, lot, expiry, storage, order and patient/procedure predicates pass.

Auditable absence. Expected receipt, scan, temperature or recall acknowledgment absent.

Metrics. Expired/incorrect use; recall time; stockouts; auditability.

Evidence: C0 R6 R26 R27

LOG-09

Returns, repair and circular-economy lineage

Near-term pilot

Addition.

Role. Preserve component lineage, condition events, repair branches and material provenance to route reuse, refurbishment or recycling.

Guard. Disposition admitted when identity, condition, warranty, safety and material predicates pass.

Auditable absence. Expected inspection or test result absent.

Metrics. Recovered value; waste avoided; turnaround; traceability.

Evidence: C0 R26 R28

LOG-10

Supply-chain risk and counterfactual branch planner

Research pilot

Addition.

Role. Use isolated branches sharing a base lineage to ask next disruption, substitute compatibility and downstream impact.

Guard. Supplier, lane, inventory or regulatory relation crosses a disruption guard.

Auditable absence. Expected shipment or supplier confirmation absent under a known reporting contract.

Metrics. Time to detect; recovery time; substitution success; scenario explainability.

Evidence: C0 R26 R27

Logistics kill criterion

If sensors cannot provide reliable coverage, identity resolution is weak or exceptions dominate, the event substrate may create false precision. Begin in one instrumented zone with a matched WMS/WES reference and explicit unknown states.

11. Smart cities and municipal governance

A municipal state substrate must make legal authority, service obligations, privacy, equity, appeal and retention as explicit as sensor geometry.

Rights-preserving smart city and municipal governance

A city query substrate should make authority, privacy, service obligations and uncertainty as explicit as geometry

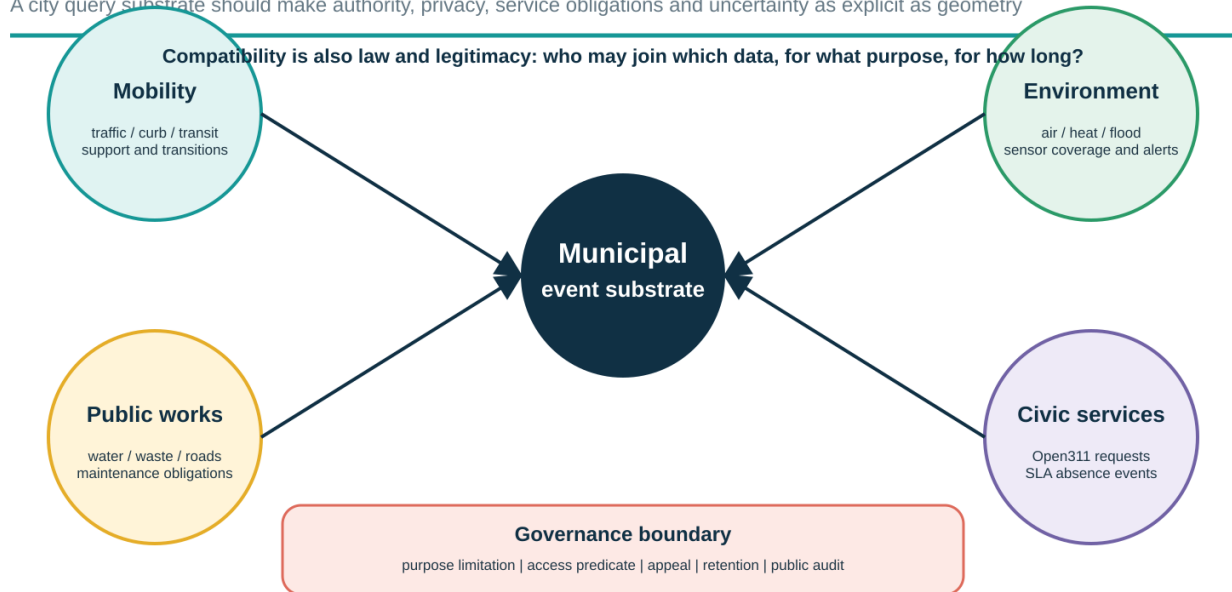


Figure 11. Compatibility in civic systems includes purpose, legal authority, access and due process - not only technical protocol matching.

NIST smart-city frameworks, KPI work, Open311 and OGC SensorThings provide existing interoperability and service interfaces. The atlas proposes a rights-preserving event layer beneath them. [R21](#) [R22](#)

[R23](#) [R24](#) [R25](#) [R35](#)

Service obligations as event contracts

A pothole report, missed transit run, water outage, waste pickup, inspection, permit or emergency handoff can create an ExpectedEvent with a due interval. Closure is not a button click; it is a verified event with support, evidence and lineage. An overdue service becomes an AbsenceEvent only when the obligation and observation coverage are valid.

CIV-01

Municipal service obligation and 311 event graph

Implementable pilot

Addition beneath CRM/asset systems.

Role. Link requests, assets, departments, contractors, SLAs, inspections and resident communications through events and auditable absences.

Guard. Request, dispatch, work, inspection and closure are typed events with reason and evidence.

Auditable absence. Expected response, visit or resolution absent after its service window.

Metrics. Resolution time; reopen rate; equity; evidence completeness; resident trust.

Evidence: C0 R21 R23 R25

CIV-02

Transit arrival and missed-service ledger

Near-term pilot

Addition.

Role. Represent scheduled/expected service, vehicle lineage, stop support and observed arrivals; emit absence only with coverage.

Guard. Arrival/departure crosses stop/time support under a vehicle/run identity.

Auditable absence. Scheduled trip absent when AVL and operational coverage are verified; otherwise unknown/data gap.

Metrics. On-time performance; ghost-trip classification; passenger delay; data coverage.

Evidence: C0 R21 R22 R24

CIV-03

Traffic, curb and intersection event control

Research pilot

Addition.

Role. Use local spatial supports, vehicle/pedestrian compatibility and next-conflict events; preserve policy versions and human override.

Guard. Conflict, queue, pedestrian call, emergency preemption or curb-occupancy guard crossing.

Auditable absence. Expected detector or phase acknowledgment absent.

Metrics. Delay; safety surrogate; emissions; false calls; accessibility.

Evidence: C0 R21 R24 R33

CIV-04

Water, sewer, waste and street-asset lifecycle

Near-term integration

Addition.

Role. Give each asset lineage, condition events, local support and maintenance guards; connect physical and budget records.

Guard. Inspection, fault, maintenance, replacement and service events update asset state.

Auditable absence. Expected inspection, collection or repair absent under a service obligation.

Metrics. Failure rate; backlog; lifecycle cost; service equity; auditability.

Evidence: C0 R21 R22 R24 R28

CIV-05**Emergency response common event substrate**

Research/operations pilot

Addition.

Role. Use a common incident lineage, local support, compatibility and event guards while preserving agency-specific sectors and permissions.

Guard. Incident, dispatch, arrival, hazard, capacity and handoff events are admitted under role and jurisdiction predicates.

Auditable absence. Expected unit acknowledgment or arrival absent within an explicit response window.

Metrics. Response time; duplicate dispatch; handoff quality; privacy incidents.

Evidence: C0 R21 R24 R35

CIV-06**Permitting, licensing and inspection event pipeline**

Implementable pilot

Addition/partial replacement of opaque workflow tracking.

Role. Model application, evidence, review, inspection, decision, appeal and policy version as events with explicit compatibility and reason codes.

Guard. Decision transition admitted only when required evidence, authority, policy version and procedural rights are satisfied.

Auditable absence. Expected review or applicant response absent after due interval, with waiver/extension states.

Metrics. Cycle time; rework; appeals; missing documents; transparency.

Evidence: C0 R28

CIV-07**Budget, procurement and contract lineage**

Implementable information-system pilot

Addition.

Role. Create lineage from policy goal to appropriation, tender, bid, award, change order, invoice, delivery and outcome.

Guard. Financial or contractual transition carries authority, policy, evidence and before/after state.

Auditable absence. Expected deliverable, report or payment evidence absent.

Metrics. Audit time; change-order visibility; duplicate spend; outcome linkage.

Evidence: C0 R28 R29

CIV-08**Policy simulation with isolated counterfactual branches**

Near-term planning pilot

Addition.

Role. Use branch-isolated policy worlds sharing base data; compare next events, service KPIs and distributional impacts without overwriting live state.

Guard. Policy-specific service, cost, risk or equity guard crossing.

Auditable absence. Missing evidence is shown as uncertainty, not zero impact.

Metrics. Scenario reproducibility; runtime; KPI coverage; distributional impact; explainability.

Evidence: C0 R22 R25

CIV-09

Rights-preserving municipal data spaces

Implementable architecture pilot

Addition and governance foundation.

Role. Use relational exclusion, purpose/time/role compatibility, provenance and public reason codes; co-location in a platform never implies join permission.

Guard. Data join admitted only for declared purpose, authority, minimization, retention and consent/legal basis.

Auditable absence. Expected deletion, review or access-log event absent triggers governance action.

Metrics. Unauthorized joins; purpose violations; access audit; resident redress.

Evidence: C0 R21 R35

CIV-10

Community well-being and service-outcome atlas

Near-term analytics pilot

Addition.

Role. Map technology events through infrastructure services to community benefits, with lineage and neighborhood support rather than one global score.

Guard. Service outcome or disparity crosses a declared community guard.

Auditable absence. Expected service result absent, with data coverage and denominator explicit.

Metrics. Service access; quality of life; equity; resilience; data completeness.

Evidence: C0 R22 R25

Governance guardrails

Required

- purpose limitation and predicate version;
- minimum necessary data and retention;
- public explanation and audit trail;
- correction and appeal;
- equity and service-outcome metrics;
- counterfactual branches isolated from production.

Prohibited

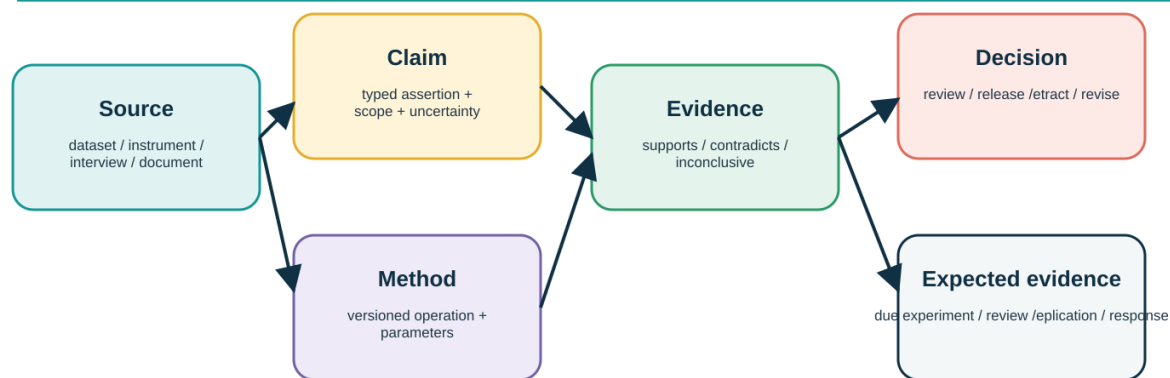
- social scoring or hidden eligibility rules;
- predictive policing from co-location or association alone;
- automated benefit denial from missing data;
- mental-health inference from civic behavior;
- permanent identity fusion across unrelated services;
- opaque optimization that removes democratic authority.

12. Research and documentation management

The strongest near-term use may be to make claims, evidence, transformations, decisions, expected reviews and missing results executable and auditable.

Research and documentation management as executable provenance

Documents remain human views; claims, evidence, transformations, decisions and gaps form the authoritative graph



Materialized views

paper, PDF, dashboard, registry entry and literature review are generated from the provenance graph; staleness and omissions are explicit.

Figure 12. Reports and dashboards become materialized views over a claim/evidence/provenance graph.

W3C PROV and FAIR principles provide established provenance and research-data anchors. The atlas adds event guards, expectations, negative results, support/compatibility and branch semantics. **R28**

R29

From document history to knowledge lineage

- Represent a claim with scope, uncertainty, evidence links and predicate version.
- Record method execution, environment, data transformation and reviewer decision as events.
- Preserve contradiction, retraction, supersession and branch history rather than overwriting.
- Create ExpectedEvidence records for review, replication, response or follow-up.
- Emit negative-result or non-observation events only when the method and observation support were valid.
- Generate papers, PDFs, dashboards and registers as views, with staleness and omitted branches declared.

RES-01

Claim-evidence-provenance graph

Implementable pilot

Addition beneath document systems.

Role. Represent claims, evidence, analyses, authorship, review and revision as a provenance graph; documents become materialized views.

Guard. Claim created, supported, challenged, revised or withdrawn with source and method lineage.

Auditable absence. Expected evidence, review or replication absent is explicit.

Metrics. Traceability; review time; unsupported claims; reproducibility.

Evidence: C0 R28 R29

RES-02

Negative-result and non-observation registry

Implementable design pattern

Addition.

Role. Use ExpectedEvent/AbsenceEvent with protocol, detection support, power, deadline and reason to record meaningful non-observation.

Guard. A pre-registered expected observation expires under adequate detection support without a compatible event.

Auditable absence. This is the core absence use case.

Metrics. Publication bias; repeat failures; power/coverage quality; reuse.

Evidence: C0 R29

RES-03

Electronic lab notebook as event-sourced experiment

Implementable pilot

Addition/partial replacement of page-centric ELN history.

Role. Record protocol, material, instrument, operator, environment, observation and transformation events with lineage.

Guard. Experiment transition emits before/after state, instrument calibration, sample lineage and confidence.

Auditable absence. Expected run, calibration or sample event absent.

Metrics. Reproducibility; audit time; missing metadata; onboarding.

Evidence: C0 R28 R29

RES-04

Living systematic review and evidence map

Near-term software pilot

Addition.

Role. Maintain study, population, intervention, outcome, bias and claim events; materialize current reviews by query.

Guard. Study or result changes the support for a claim under a versioned inclusion predicate.

Auditable absence. Expected registry result or follow-up publication absent is tracked as pending/unknown.

Metrics. Update latency; citation coverage; reproducibility; conflict visibility.

Evidence: C0 R28 R29

RES-05

Regulatory and quality-management evidence bundle

Implementable integration

Addition.

Role. Use event lineage and compatibility to generate auditable, version-specific evidence bundles.

Guard. Requirement, design, test, deviation, risk control and approval events link to a release.

Auditable absence. Expected verification or approval absent blocks release.

Metrics. Audit prep time; trace gaps; change impact; defect escape.

Evidence: C0 R28

RES-06

Software and computational-environment lineage

Implementable pilot

Addition.

Role. Treat environment and execution as lineage events; reconstruct exact branch and artifacts for each result.

Guard. Run event links code commit, dependencies, data, hardware, parameters and outputs.

Auditable absence. Expected artifact or checksum absent.

Metrics. Reproduction success; build time; provenance completeness.

Evidence: C0 R28 R29

RES-07

Research portfolio and funding lineage

Near-term information-system pilot

Addition.

Role. Build lineage from question and funding to experiments, assets, outputs, reuse and societal outcomes without collapsing evaluation into one score.

Guard. Funding, milestone, dataset, publication, reuse and outcome events preserve attribution.

Auditable absence. Expected deliverable or data release absent with documented reason.

Metrics. Duplication; reuse; time to evidence; administrative burden.

Evidence: C0 R28 R29

RES-08

Knowledge transfer and institutional memory substrate

Near-term pilot

Addition.

Role. Store decisions, assumptions, predicates, evidence and outcomes as lineage-aware events; documents and dashboards are generated views.

Guard. Decision or rule change records context, authority, evidence, alternatives and consequences.

Auditable absence. Expected review or retirement event absent.

Metrics. Onboarding time; repeated mistakes; decision traceability; stale rules.

Evidence: C0 R28

Documentation management impact

The architecture can reduce repeated manual reconciliation among reports, tickets, spreadsheets, notebooks, regulations and dashboards. Its value is not automatic truth; it is explicit lineage and a smaller gap between what a document says, what evidence supports it and what event changed it.

13. Pilot portfolio and build order

Start with headless, bounded experiments that can fail usefully before attempting new instruments or large institutional transformation.

Recommended sequence

1. **Freeze schemas.** Use the included State, Event, ExpectedEvent, AbsenceEvent, Compatibility and Lineage schemas.
2. **Build Equation World Zero.** Implement state-at-time, next-event, events-in-support, can-couple, transition routing and identity reconstruction on one relation family.
3. **Deploy auditable absence.** Choose one low-risk workflow such as a research review, warehouse scan or municipal SLA.
4. **Add one domain twin.** MRI replay, PV array, fab chamber, pipeline test loop or hospital care journey.
5. **Compare with a locked baseline.** Equal task, error, uncertainty, coverage and safety.
6. **Only then add analog/photonic hardware.** Include full source, conversion, control and calibration costs.

20. Risks, failure modes, and kill criteria

The mature corpus is strongest where it admits failure. Long horizons, logarithmic coordinates, phase schedules, singular centers, near-singular operators, tangencies, coincident surfaces, rank changes, branch explosion, event explosion, and semantic grounding are not implementation footnotes. They determine whether the ontology can be executed at all. [S1 pp. 8, 11]

Risk	Failure symptom	Required response	Kill criterion
Algebraic closure	Composition leaves the solvable/normalized relation family.	Restrict grammar, normalize, or compile to certified approximation.	Expression growth eliminates direct-query advantage.
Event explosion	Too many roots or candidate relations.	Improve support/compatibility, bound interval, simplify relation family.	Solver work approaches/exceeds materialization baseline.
Degeneracy	Tangency, coincident surfaces, multiple roots, unstable ordering.	Interval methods, multiplicity tracking, tie policy, uncertainty.	Missed or unordered events exceed tolerance.
Numerical conditioning	Long-horizon phase drift, log singularity, rank instability.	Scaled coordinates, certified numerics, error bounds, reference solver.	Answers cannot be certified at required tolerance.
Branch explosion	Counterfactual history grows combinatorially.	Prune, merge with explicit policy, summarize, checkpoint.	Memory or query cost scales with all unbounded branches.
Lineage ambiguity	Splits/merges cannot be reconstructed.	Explicit parent sets and reconciliation rules.	Identity collisions or unrecoverable ancestry.
Semantic underspecification	Compatibility says "yes/no" without explainable policy.	Reason codes, versioned predicates, typed permissions.	System cannot audit why an event was admitted.
Projection feedback	Display artifacts alter authoritative state.	One-way view contract; explicit sensor-to-log events only.	Framebuffer becomes implicit state store.
Hardware drift	Optical transfer changes with temperature/aging/liquid state.	Reference channels, calibration, bounded uncertainty.	Calibration energy/time erases benefit.

Source report page 41: The source risk and kill-criteria page makes ordinary engineering, matched baselines and falsifiability mandatory.

Sixteen bounded pilots

P01. Equation World Zero

Digital core | 3-6 months

Implement six queries on one relation family and benchmark against frame stepping.

Gate: Go if event solving + lineage is cheaper at equal correctness.

P02. Auditable absence ledger

Cross-domain data | 3 months

Deploy ExpectedEvent/AbsenceEvent in hospital follow-up, shipment scan or municipal SLA.

Gate: Go if unknown vs absent classification improves without excessive false absence.

P03. MRI adaptive raw-data replay

Biomedical imaging | 6-12 months

Replay archived k-space/physiology data to test support- and guard-driven acquisition decisions.

Gate: Go only with preserved task performance and uncertainty.

P04. Quantitative MR direct query phantom

Biomedical imaging | 9-18 months

Compare direct parameter estimation, MRF-style approach and image-first baseline.

Gate: Go if task accuracy/scan time improve under safety constraints.

P05. Cradle-to-life care journey

Health information | 6-12 months

FHIR/PROV event graph for maternal-neonatal or chronic-care pathway.

Gate: Go if identity, consent and care-gap auditability improve.

P06. Mental-health patient-controlled registry

Mental health | 12 months

Longitudinal goals, context, consent and research measures with no automated diagnosis.

Gate: Go only with independent ethics, patient control and low harm.

P07. Lithography hotspot event solver

Semiconductor | 6-12 months

Local support and guard queries on a synthetic/public mask benchmark.

Gate: Go if hotspot recall and runtime beat baseline without missed long-range effects.

P08. Fab next-fault shadow twin

Manufacturing | 6-12 months

One chamber subsystem with continuous relations and event predictions.

Gate: Go if false alarms and lead time meet preset thresholds.

P09. Hollowlens metrology proxy

Physical/photonic | 12-24 months

Optofluidic or fixed photonic prefilter plus digital sidecar and matched electronic baseline.

Gate: Go only on end-to-end verified events/J and stability.

P10. PV array event twin

Solar | 6-12 months

String-level faults, absence semantics and O&M workflow.

Gate: Go if energy recovery exceeds sensor/control burden.

P11. Pipeline leak replay and test loop

Infrastructure | 9-18 months

Historical transients plus controlled leak/negative-pressure-wave tests.

Gate: Go only with regulatory-grade sensitivity and false-alarm bounds.

P12. Warehouse event truth zone

Logistics | 6 months

EPCIS lineage, expected scans and AMR time-of-conflict in one zone.

Gate: Go if inventory accuracy and throughput improve.

P13. Municipal service obligation graph

Civic | 6 months

Pothole, streetlight or waste service with Open311 and explicit SLA absence.

Gate: Go if closure evidence and equity improve.

P14. Research claim-evidence graph

Documentation | 3-6 months

Convert one report or systematic review to PROV/FAIR event graph.

Gate: Go if review/update and audit time fall.

P15. Event ADC sensing benchmark

Mixed signal | 6-12 months

Level-crossing plus reference sampling on vibration, ECG or environmental signal.

Gate: Go if end-to-end energy falls at equal reconstruction/event performance.

P16. Photonic query-kernel benchmark

Compute hardware | 12-24 months

Stable transform with full source, detector, ADC/DAC and calibration accounting.

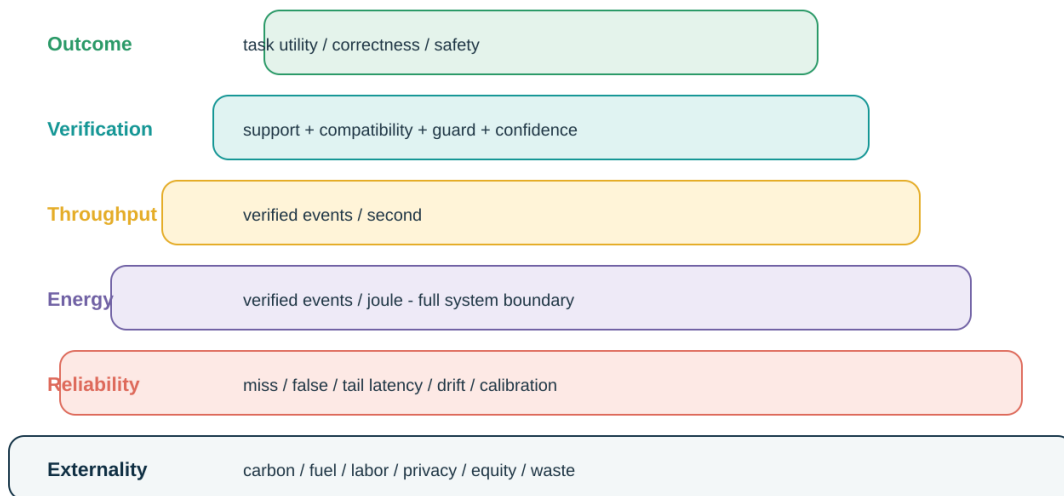
Gate: Go only if end-to-end J/query beats electronic reference.

14. Throughput, joules and benchmark contract

Count verified outcomes at a declared boundary, not photons, raw messages, pixels, arithmetic operations or theoretical propagation speed.

Benchmark stack: throughput means verified outcomes at a declared

Do not count propagation rhetoric, raw detector traffic or unpriced compute as system advantage



Every application must beat a matched baseline at equal error, uncertainty and safety - or it remains a useful negative result.

Figure 13. System advantage is measured from outcome and verification through energy, reliability and externalities.

$$\text{Theta_verified} = N_verified / \Delta t$$

A verified event satisfies support, compatibility, guard and confidence.

$$\text{Theta_E} = N_verified / E_total$$

E_total includes sensing/source, conversion, actuation, compute, control, storage, communications and calibration.

Minimum comparison protocol

1. Pre-register the task, support, compatibility predicate, guard, uncertainty and failure thresholds.
2. Lock the electronic, clinical, operational or physics baseline before inspecting results.
3. Measure candidate counts before and after support and compatibility pruning.
4. Report miss, false-event, ordering, tail latency, drift and recalibration burden.
5. Include every system energy component and amortized hardware/control overhead.
6. Report task outcome, not only internal throughput.

7. Record where the method falls back to sampling, dense solving or manual review.
8. Stop or narrow the claim when the baseline wins.

Core metrics

Metric	Unit / definition	Interpretation
Theta_verified	verified events / second	Events satisfying support, compatibility, guard and confidence.
Theta_E	verified events / joule	Includes sensing/source, conversion, actuation, control, calibration and storage.
eta_route	verified / admitted	Correct routing among admitted candidates.
R_support	all candidates / supported candidates	Pruning due to local radial/angular/causal/semantic support.
R_compat	supported / solver-entering candidates	Pruning due to physical, policy, identity and mode compatibility.
L50	median event latency	Median time from physical or digital condition to verified event.
L99	99th percentile event latency	Tail latency including settling, network and control.
P_miss	missed-event probability	Probability of failing to emit a required event.
P_false	false-event probability	Probability of emitting an unjustified event.
P_abs_false	false-absence probability	Absence emitted despite an unobserved or late compatible event.
P_unknown	unknown/missing-coverage fraction	Cases that cannot be classified as present or absent.
coverage_support	time/space within certified observation support	Required for meaningful negative events.
lineage_completeness	reconstructable transitions / required transitions	Coverage of parent/child, split/merge and provenance edges.
replay_fidelity	reconstructed state matches reference	Correctness of seed+grammar+log reconstruction.
branch_growth	active branch nodes / scenario	Counterfactual growth and pruning burden.
expression_growth	relation complexity vs grammar depth	Algebraic closure and normalization burden.
solver_work	root/constraint operations per verified event	Core event-solver cost.
materialization_avoided	bytes or operations not materialized	State/view work avoided by direct queries.

Energy and carbon reporting

Events per joule is a systems metric, not automatically a climate claim. For energy, manufacturing or compute applications, report the measured energy delta and separately convert it into fuel, emissions, cost or capacity impacts with transparent assumptions.

15. Risks, prohibited shortcuts and kill criteria

The architecture should be judged by the points where it can fail: algebraic closure, event density, coverage, conditioning, governance and full-system economics.

Technical failure modes

- **Expression or grammar explosion:** direct queries become larger than the state they avoid materializing.
- **Event explosion and degeneracy:** dense roots, tangencies, simultaneous guards and uncertain ordering erase the advantage.
- **Weak support or compatibility:** filters fail to prune, turning a local query into a global scan.
- **Numerical instability:** long horizons, singularities, near-coincident surfaces and model mismatch make roots unreliable.
- **Branch and lineage explosion:** splits, merges and counterfactuals become ad hoc bookkeeping.
- **Coverage ambiguity:** absent observation is mislabeled as negative evidence.
- **Calibration burden:** analog, photonic or sensor drift consumes the energy/time benefit.
- **Sidecar dominance:** a conventional digital system performs nearly all useful work.
- **Institutional mismatch:** workflow, staffing, regulation or legal authority prevents the event model from producing real outcomes.

Safety, ethics and rights failure modes

- A medical or mental-health predicate becomes an autonomous decision-maker.
- Identity lineage becomes surveillance or permanent cross-context linkage.
- Compatibility becomes an opaque access or eligibility rule without appeal.
- Municipal optimization outruns democratic authority or concentrates harm.
- Absence events punish people for sensor failure, digital exclusion, opt-out or transfer.
- A physical metaphor is used to claim immunity, biological control or impossible containment.

Universal kill criteria

Stop, narrow or retain the baseline when

- task correctness or safety is worse;
- uncertainty is uncalibrated;
- event ordering cannot be certified within tolerance;
- full-system events/J does not improve;
- calibration or governance burden erases the benefit;
- support and compatibility do not stay sparse;
- the claimed absence cannot be distinguished from missing observation;
- the application requires a totalizing claim the corpus itself rejects.

16. Recommended program architecture

Organize the work as a reusable kernel, domain adapters, evidence contracts and matched reference systems.

Four-layer implementation

1. Core kernel

State evaluator, support planner, compatibility engine, certified event solver, transition router, lineage graph and expectation/absence ledger.

2. Domain adapter

MRI physics, lithography models, grid equations, pipeline monitoring, hospital FHIR workflows, logistics EPCIS, municipal service rules or research PROV.

3. Projection and control adapters

Images, dashboards, reports, alarms, robot commands, inverter commands, scanner controls and human review interfaces.

4. Evidence and governance

Baseline, validation, uncertainty, calibration, energy boundary, access/consent, audit, appeal, kill criteria and change control.

Recommended first year

- Months 0-3: run the included software prototype and deploy an auditable-absence pilot in a non-clinical workflow.
- Months 3-6: implement Equation World Zero and one data-pipeline or research-provenance adapter.
- Months 6-12: select two domain pilots - one physical/industrial and one human/institutional - with external subject-matter owners.
- Months 9-18: pursue MRI replay, lithography hotspot, PV/pipeline or event-ADC benchmarks only after schemas and baseline discipline are stable.
- Months 12-24: consider Hollowlens or photonic kernels if the digital model shows a stable, narrow transfer function worth physicalizing.

Program-level success

Success is not a picture of a new world. It is a repeatable family of bounded applications in which direct state/event queries, auditable absence and lineage provide better outcomes with lower materialization cost at a declared uncertainty and governance boundary.

Full 95-application atlas

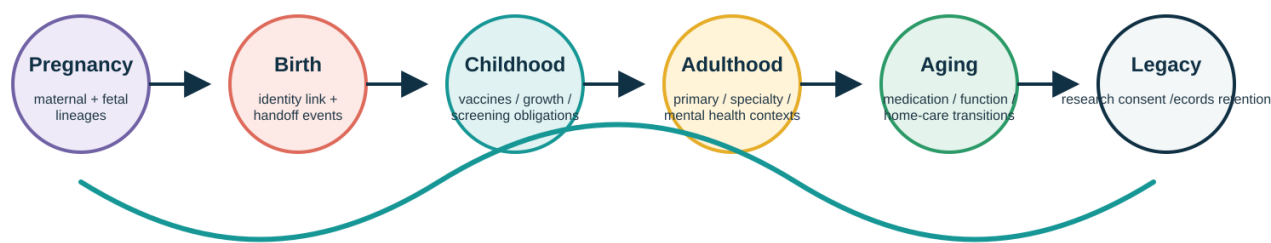
Each card records the proposed role, boundary, event guard, auditable absence, metrics and evidence links.

The cards are hypotheses and pilot definitions. Maturity labels refer to implementation readiness of the information architecture, not proof of domain performance.

Biomedical & healthcare

Cradle-to-life health journey as a consented event and lineage graph

A person is not a snapshot; care episodes, measurements, decisions and absences remain linked without becoming the person



Care gaps become auditable
Only under valid obligation and coverage.

Labels remain projections
Diagnosis, risk score and note never exhaust identity.

Research reuse is gated
Purpose, consent, provenance and revocation are explicit.

Application overview: Biomedical & healthcare

BIO-01 MRI adaptive acquisition and sequence orchestration

Near-term research pilot

Addition to MRI acquisition and control; not a magnet replacement.

Role. Use local anatomical/physiological support, motion and safety compatibility, and quantitative guard conditions to adapt which measurements are acquired next.

Guard. Acquire the next encoding block when predicted information gain or quantitative uncertainty crosses a declared threshold and safety constraints remain satisfied.

Auditable absence. Expected calibration, physiological gating or coil data fail to arrive within the sequence window despite verified sensor coverage.

Metrics. Diagnostic task performance; scan time; acquired samples; SAR and gradient limits; motion artifacts; events/j; false-adaptation rate.

Evidence: C0 R1 R2 R3 R5

BIO-02 MRI physics-constrained direct parameter query

Research prototype

Partial replacement of selected reconstruction stages for bounded tasks.

Role. Represent Bloch-model-compatible signal families as relations and query selected parameters or change events directly, optionally using images as a view.

Guard. Parameter posterior or certified interval enters a clinically meaningful range under sufficient acquisition support.

Auditable absence. A parameter cannot be certified because the required excitation/encoding support was never observed.

Metrics. Parameter bias/variance; calibration; scan time; task AUC; reconstruction energy; uncertainty coverage.

Evidence: C0 R3 R4

BIO-03

MRI narrow diagnostic instrument as an alternative to full imaging

Frontier concept

Alternative narrow instrument; full MRI replacement unsupported.

Role. Design a task-specific MR instrument that answers a bounded question such as relaxation change, flow threshold or lesion-presence probability without requiring a full diagnostic image.

Guard. A task-specific signal relation crosses a validated threshold with confidence and safety conditions satisfied.

Auditable absence. The expected signal support is absent, but only after coil, sequence and coverage integrity are verified.

Metrics. Sensitivity/specificity; acquisition time; energy; footprint; cost; inconclusive rate; referral-to-full-MRI rate.

Evidence: C0 R1 R4 R5

BIO-04

MRI motion, respiratory and cardiac event gating

Near-term pilot

Addition.

Role. Treat motion/physiology as continuous state and solve for compatible acquisition windows or correction events rather than replaying uniform frames.

Guard. Predicted motion enters an admissible window and a sequence block is admitted; otherwise hold or redirect.

Auditable absence. A required quiet-window event never occurs within the allowed interval.

Metrics. Rejected acquisitions; reacquisition rate; image/parameter quality; latency; patient time.

Evidence: C0 R1 R3

BIO-05

MRI safety and implant compatibility engine

Implementable integration

Addition to safety workflow.

Role. Model implant identity, field strength, sequence limits and patient state as a compatibility predicate with provenance and explicit reason codes.

Guard. Scan becomes admissible only when all required device attributes and MR-conditional constraints are present and current.

Auditable absence. Missing implant identity or condition is logged as a blocking absence, not silently treated as safe.

Metrics. Screening completeness; prevented unsafe scans; delay time; override rate; audibility.

Evidence: C0 R5 R6

BIO-06

Ultrasound echo-event acquisition and beam control

Research prototype

Addition or partial replacement of acquisition/reconstruction for narrow tasks.

Role. Use level-crossing or feature-triggered acquisition, local beam support and direct arrival-time/flow events to reduce irrelevant conversion and processing.

Guard. Echo arrival, Doppler shift or tissue-boundary relation crosses a validated threshold within the beam support.

Auditable absence. Expected echo absent only when transducer contact and acoustic coverage are verified.

Metrics. Samples/event; energy; latency; sensitivity; localization error; image equivalence when a view is produced.

Evidence: C0 R31

BIO-07**CT/PET task-specific sparse acquisition**

Research concept

Addition/partial replacement in bounded tasks.

Role. Select measurement supports adaptively and solve for bounded findings or quantitative changes with explicit uncertainty, retaining full reconstruction as fallback.

Guard. Task confidence reaches a preset boundary without violating dose/count constraints.

Auditable absence. A required angular or count support is absent; result remains inconclusive rather than negative.

Metrics. Dose or tracer counts; task performance; scan time; uncertainty; fallback rate.

Evidence: C0

BIO-08**Microscopy and cell-culture event microscope**

Near-term research pilot

Addition and, for narrow assays, partial replacement of continuous imaging.

Role. Use local optical supports, event sensors or analog prefilters to emit only lineage-aware cell events such as division, migration, morphology change or fluorescence threshold crossing.

Guard. A tracked cell lineage crosses a morphology, intensity or interaction guard with sufficient optical support.

Auditable absence. Expected cell or marker observation absent only when field coverage and sensor health are known.

Metrics. Events/J; bytes/experiment; missed/false events; lineage accuracy; phototoxic exposure.

Evidence: C0 R30

BIO-09**ICU and ward event-native patient monitoring**

Near-term pilot

Addition to monitoring and clinical decision support.

Role. Maintain a continuous patient state, local physiological supports, compatibility with interventions and an auditable event record for threshold crossings and trends.

Guard. A validated physiological relation crosses a severity guard and matches the patient context, sensor quality and treatment phase.

Auditable absence. Expected vital, medication response or reassessment is absent despite confirmed observation coverage.

Metrics. Alarm precision/recall; time to recognition; clinician workload; missed-data rate; intervention outcomes.

Evidence: C0 R6 R7 R8 R10

BIO-10**Surgical navigation and tool-tissue time-of-contact**

Research prototype

Addition.

Role. Represent tool and anatomy trajectories continuously; solve certified time-of-contact events within local support and surgical permissions.

Guard. Earliest predicted tool/tissue contact or no-go-zone crossing under uncertainty.

Auditable absence. Tracking signal absence blocks the action rather than implying clearance.

Metrics. Contact-time error; missed crossings; latency; false stops; surgeon workload.

Evidence: C0 R33 R34

BIO-11

Medication, infusion and closed-loop therapy guard engine

Near-term pilot

Addition.

Role. Join medication identity, dose lineage, patient state, timing and device status through compatibility and event guards; keep clinicians in the loop.

Guard. Dose or infusion transition is admitted only when patient, order, pump, drug, route and timing are compatible.

Auditable absence. Expected reassessment, pump acknowledgment or lab result is missing after its due interval.

Metrics. Medication errors; delayed doses; false blocks; response-monitoring completeness; audit time.

Evidence: C0 R6 R7 R8

BIO-12

Cradle-to-life longitudinal patient journey substrate

Implementable architecture pilot

Addition beneath EHRs; not an autonomous medical authority.

Role. Use a stable person lineage with prenatal, birth, neonatal, childhood, adult and elder branches; represent encounters and documents as views over a consent- and provenance-aware event graph.

Guard. Care transitions, referrals, procedures, consent changes and outcomes are typed events tied to the person lineage.

Auditable absence. Expected follow-up, screening, handoff, result or support service is absent only under an explicit obligation and coverage rule.

Metrics. Record-linkage errors; care-gap closure; duplicate testing; handoff failures; patient access; consent compliance.

Evidence: C0 R6 R7 R8 R28

BIO-13

Maternal, fetal, birth and neonatal journey coordination

Near-term information-system pilot

Addition.

Role. Model mother, fetus/newborn, samples, devices and care teams as related but distinct lineages, with consent and event transitions at birth.

Guard. Birth, transfer, screening, feeding, medication and follow-up transitions preserve parent/child lineage and provenance.

Auditable absence. Expected screening or post-discharge follow-up is absent with verified obligation and outreach coverage.

Metrics. Missed screens; duplicate identity; handoff delay; maternal/newborn linkage accuracy.

Evidence: C0 R6 R8

BIO-14

Hospital flow, beds, theatres and discharge event twin

Near-term pilot

Addition/partial replacement of dashboard-centric coordination.

Role. Represent patient flow, rooms, staff, equipment and constraints as a compatibility-gated event system with next-event and counterfactual queries.

Guard. A transfer or discharge becomes admissible when clinical, staffing, bed, transport and documentation conditions are satisfied.

Auditable absence. Required consult, result, transport or discharge action has not occurred by the due interval.

Metrics. Length of stay; boarding time; cancelled procedures; idle beds; false-ready rate; staff burden.

Evidence: C0 R6 R7 R10

BIO-15

Mental-health longitudinal care and research substrate

Research and governance pilot

Addition to records and research; not automated diagnosis or therapy.

Role. Maintain patient-authored goals, consent, context, development, environment, observations and interventions as lineage-aware events and dimensional trajectories.

Guard. A care or research transition is admitted under explicit consent, role, time and safety predicates, with human review.

Auditable absence. Expected check-in or follow-up may be logged as absent, but lack of digital activity is never equated with deterioration without corroboration.

Metrics. Continuity; patient control; follow-up completion; bias; clinician burden; data access.

Evidence: C0 R8 R9 R10 R11

BIO-16

Clinical trial and observational study event graph

Implementable pilot

Addition/partial replacement of document-centric trial operations.

Role. Create a versioned event and provenance graph connecting protocol, consent, intervention, measurement, deviation, analysis and publication.

Guard. Every trial action creates a typed event with protocol version, authority, subject, time and before/after state.

Auditable absence. Expected visit, sample, outcome or adverse-event assessment is explicitly expired, waived or observed.

Metrics. Protocol deviations; reconciliation time; missing data; audit readiness; reproducibility.

Evidence: C0 R6 R28 R29

BIO-17

Biomedical sample and biobank lineage

Implementable integration

Addition.

Role. Use lineage addresses and split/merge rules for specimen, aliquot, derivative, assay and storage events.

Guard. A sample transition is admitted only when identity, consent, chain of custody, storage and assay compatibility are satisfied.

Auditable absence. Expected receipt, temperature record or assay result is missing after verified custody.

Metrics. Misidentification; unusable samples; audit time; consent violations; freezer excursions.

Evidence: C0 R26 R27 R28

BIO-18

Protein dynamics and molecular-event simulation

Research prototype

Hybrid addition; broad replacement unsupported.

Role. For selected relation families, solve binding, contact, conformational and threshold events directly, while retaining conventional simulation for unresolved regions.

Guard. Earliest compatible contact, barrier crossing or conformational transition within a bounded model.

Auditable absence. A predicted event is absent only when the modeled support and force field are valid; otherwise unknown.

Metrics. Event-time error; energy conservation; missed states; compute/J; expression growth.

Evidence: C0

BIO-19

CRISPR and gene-editing in-silico candidate pruning

Research concept**Addition to design; not physical gene control.**

Role. Use local genomic/epigenetic support, compatibility predicates and lineage to rank candidate interventions and downstream event risks before laboratory testing.

Guard. Candidate passes sequence, structure, cell-context, off-target and policy guards with uncertainty.

Auditable absence. Missing cell-context evidence is a blocking absence, not proof of safety.

Metrics. Off-target ranking; experimental hit rate; uncertainty calibration; provenance.

Evidence: C0 R8 R9

BIO-20

Public-health event and service-absence observatory

Near-term data pilot**Addition.**

Role. Use event notifications, local geographic supports, lineage and explicit expected-report contracts to distinguish true absence, delayed data and sensor failure.

Guard. Case, lab, wastewater, capacity or intervention events cross a public-health guard.

Auditable absence. Expected report absent only after source coverage, reporting obligation and deadline are verified.

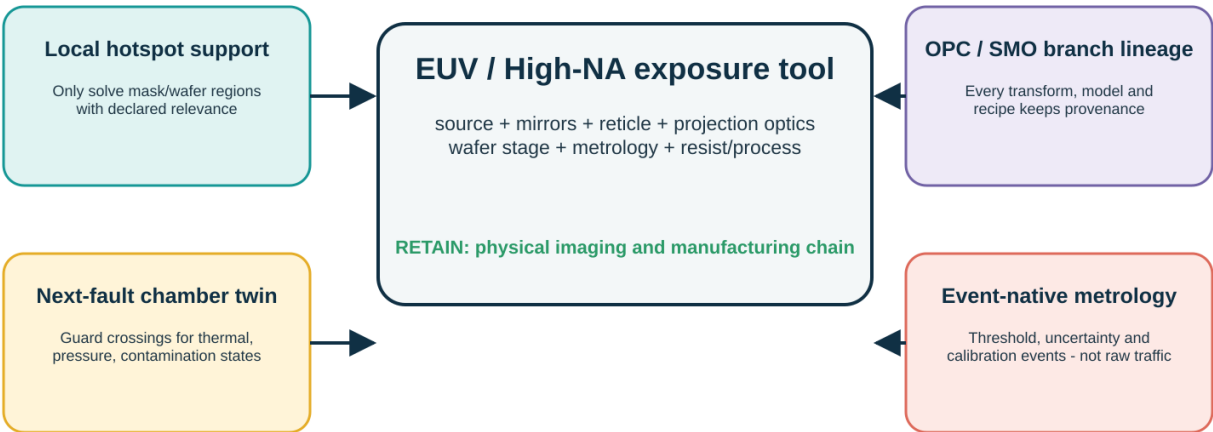
Metrics. Reporting delay; missingness classification; false alarms; privacy; action time.

Evidence: C0 R6 R7 R8 R24

Semiconductor & physical manufacturing

ASML / lithography: event-native augmentation

Change control, optimization and provenance - not the physical creation of EUV light, mirrors, stages or resist chemistry



Kill criterion: if locality, calibration and solver costs do not beat the full reference at equal wafer outcome, retain the baseline.

Application overview: Semiconductor & physical manufacturing

PHY-01 ASML/EUV event-native exposure and scanner control Research/industry pilot Addition to scanner control; physical exposure system remains. Role. Add a query-first state/event layer for wafer-stage trajectories, dose, focus, overlay, contamination and subsystem readiness. Guard. Expose only when stage, source, reticle, wafer, focus, dose and safety states are compatible; latch deviations as events. Auditable absence. Expected metrology, droplet/source, stage or vacuum readiness event is absent within the control window. Metrics. Overlay; focus/dose error; wafers/hour; downtime; false aborts; control energy. Evidence: C0 R12 R13 R14	PHY-02 Computational lithography hotspot query engine Research prototype Addition/partial replacement of selected optimization passes. Role. Represent pattern constraints and process windows as local supports and guards; query hotspot events and compatible corrections instead of materializing every candidate globally. Guard. A local pattern/process relation exits the manufacturable window under a declared model. Auditable absence. No hotspot observation is valid only under verified model coverage; unmodeled pattern classes remain unknown. Metrics. Runtime; memory; hotspot recall; process-window improvement; correction complexity. Evidence: C0 R12
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PHY-03

Reduced-order Maxwell/ rigorous simulation query service

Research prototype

Addition.

Role. Compile bounded parameter families into reduced or surrogate relation models with certified validity regions and fall back to full solvers outside support.

Guard. Parameter query is answered only within certified support and error tolerance.

Auditable absence. A result is absent/inconclusive when the query lies outside the reduced model support.

Metrics. Error bound; solver speed; fallback rate; memory; process-window coverage.

Evidence: C0 R12 R13

PHY-04

Reticle, wafer and process lineage graph

Implementable integration

Addition beneath MES/yield systems.

Role. Use immutable lineage, split/merge and event records across design, mask, exposure, etch, metrology and yield.

Guard. Every custody, recipe, exposure, inspection and disposition creates a versioned event.

Auditable absence. Expected metrology, hold release or lot movement is absent after a declared due interval.

Metrics. Root-cause time; genealogy completeness; scrap containment; auditability.

Evidence: C0 R26 R27 R28

PHY-05

Wafer-stage and robot continuous motion scheduling

Near-term research pilot

Addition.

Role. Use continuous trajectories, local supports and exact/conservative time-of-impact events rather than only fixed-rate checking.

Guard. Earliest contact, clearance or timing-constraint crossing under bounded trajectory uncertainty.

Auditable absence. Tracking absence blocks motion; it never implies free space.

Metrics. Cycle time; missed collision probability; false stops; timing jitter.

Evidence: C0 R33 R34

PHY-06

Fab chamber next-fault digital twin

Near-term pilot

Addition.

Role. Represent pressure, temperature, flow, plasma, chemistry and equipment states continuously; solve next-fault and regime-change events with calibrated uncertainty.

Guard. Earliest compatible threshold or regime transition under a validated physical model.

Auditable absence. Expected sensor or actuator acknowledgment absent with coverage known.

Metrics. Unplanned downtime; false alarms; yield; maintenance hours; model error.

Evidence: C0

PHY-07

Metrology optical prefilter and B.C.E. detector

Frontier lab prototype

Addition; narrow analog front end.

Role. Use calibrated waveguide/optofluidic transfer functions and compatibility gates to detect narrow optical events before full digitization.

Guard. Selected wavelength/polarization/mode response crosses a calibrated guard.

Auditable absence. Expected reference-channel event absent indicates drift or hardware fault, not sample absence.

Metrics. Verified events/J; insertion loss; miss/false rate; calibration interval; throughput.

Evidence: C0

PHY-08

PCB routing, revision and signal-integrity event graph

Near-term software pilot

Addition/partial replacement of selected checks.

Role. Treat traces, vias, components and constraints as lineage-aware relations; use local support for clearance and compatibility for electrical/net semantics.

Guard. Clearance, impedance, timing or net-compatibility relation crosses a guard.

Auditable absence. Expected test or sign-off absent is explicit.

Metrics. DRC runtime; missed violations; revision traceability; ECO turnaround.

Evidence: C0

PHY-09

Pick-and-place event-native cell

Near-term pilot

Addition.

Role. Combine continuous motion, local supports, compatibility and lineage for part/nozzle/feeder events.

Guard. Pick or placement is admitted when pose, part identity, nozzle, feeder, board and clearance states match.

Auditable absence. Expected component present but not observed under verified camera/feeder coverage.

Metrics. Cycle time; mispicks; placement error; stoppages; energy/placed part.

Evidence: C0 R26 R33 R34

PHY-10

Additive manufacturing melt-pool event control

Research prototype

Addition.

Role. Model melt-pool geometry, thermal state, feedstock and scan trajectory continuously; trigger corrective transitions at guard crossings.

Guard. Melt-pool, porosity-risk or thermal-gradient relation exits process support.

Auditable absence. Expected sensor or powder-delivery event absent blocks confidence.

Metrics. Defect rate; scrap; energy/kg; latency; calibration.

Evidence: C0

PHY-11

CNC and tool-wear next-event control

Near-term pilot

Addition.

Role. Represent tool state, workpiece lineage and continuous load/vibration relations; predict wear threshold and collision events.

Guard. Predicted wear, chatter, overload or collision guard crossing.

Auditable absence. Expected cut completion or sensor signature absent under verified coverage.

Metrics. Tool life; scrap; downtime; false alarms; energy/part.

Evidence: C0

PHY-12

Industrial optical spectroscopy classifier

Frontier lab pilot

Addition/alternative narrow instrument.

Role. Use tunable optical supports, mode compatibility and threshold events to detect selected chemical/physical signatures before or alongside full spectra.

Guard. Spectral/mode response crosses a calibrated target guard.

Auditable absence. Reference line absent indicates instrument fault; target absence requires full support coverage.

Metrics. Events/J; limit of detection; false alarms; calibration; cost.

Evidence: C0

PHY-13

Quality inspection as direct relation query

Near-term pilot

Addition/partial replacement for narrow inspections.

Role. Use local support and event-based or analog prefilters for specific geometry, surface or assembly relations; retain images only for exceptions.

Guard. Feature deviation crosses tolerance under sufficient illumination and pose support.

Auditable absence. Expected feature not observed under verified coverage; otherwise unknown.

Metrics. Bytes/part; events/J; miss/false; inspection time; exception-image rate.

Evidence: C0 R30

PHY-14

Photonic relation kernel for fixed industrial queries

Frontier prototype

Addition; domain-specific accelerator.

Role. Map stable linear transforms to calibrated interferometric hardware, while nonlinear logic, uncertainty, lineage and policy remain digital.

Guard. Optical output crosses a calibrated guard and digital sidecar validates context.

Auditable absence. Reference-channel absence or drift blocks result.

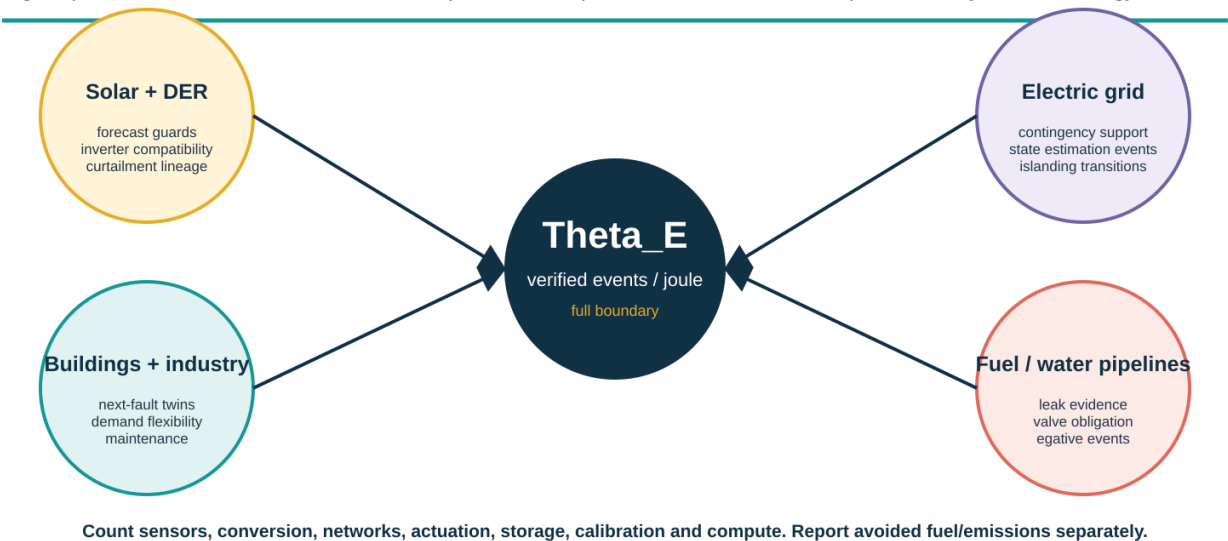
Metrics. End-to-end events/J; accuracy; calibration; insertion loss; latency.

Evidence: C0 R32

Energy, climate & infrastructure

Energy, grid, pipeline and solar application map

High impact comes from better measurement, dispatch, leak response, maintenance and compute efficiency - not free energy



Application overview: Energy, climate & infrastructure

ENG-01 Grid state and contingency query substrate Research/utility pilot Addition beneath EMS/DERMS. Role. Maintain a versioned grid state and solve local support, compatibility and next-limit events; use conventional power-flow solvers as typed physical services. Guard. Earliest thermal, voltage, frequency, stability or protection guard crossing under a scenario branch. Auditable absence. Expected telemetry or breaker state absent is classified separately from a normal zero. Metrics. Contingency solve time; missed events; false alarms; operator workload; events/]. Evidence: C0 R15 R16	ENG-02 Distributed energy and microgrid compatibility router Near-term pilot Addition. Role. Use compatibility predicates for device capability, grid state, market/policy and ownership; route only admitted events. Guard. Dispatch or islanding transition admitted when electrical, contractual, cyber and timing predicates match. Auditable absence. Expected device acknowledgment or telemetry absent after a verified communication window. Metrics. Dispatch success; interoperability failures; curtailment; resilience; cyber incidents. Evidence: C0 R15 R17
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ENG-03

Protection and disturbance event engine

Near-term research pilot**Addition to analysis and coordination, not unvalidated relay replacement.**

Role. Represent disturbance supports, device compatibility, guard crossings and event lineage explicitly; keep certified relays authoritative.

Guard. Fault or instability relation crosses a certified boundary in the protected zone.

Auditable absence. Expected breaker trip or relay acknowledgment absent within the mandated interval.

Metrics. Detection time; selectivity; false trips; coordination; event reconstruction.

Evidence: C0 R15

ENG-04

Solar PV string and module event twin

Near-term pilot**Addition.**

Role. Use local array/string support, expected-yield relations, inverter compatibility and lineage to detect shading, soiling, mismatch, connector and degradation events.

Guard. Measured/predicted energy balance exits a calibrated support or a component relation crosses a fault guard.

Auditable absence. Expected generation or telemetry absent only after irradiance, availability and sensor coverage are verified.

Metrics. Energy recovered; detection delay; false alarms; O&M cost; events/J; curtailment avoided.

Evidence: C0 R17 R18

ENG-05

Solar inverter and MPPT event-native control

Research pilot**Addition; low-level control remains conventional unless certified.**

Role. Add local event rules for mode changes, mismatch, voltage support and thermal limits; investigate level-crossing conversion for sparse supervisory signals.

Guard. Operating point, thermal, grid-support or protection relation crosses a guard.

Auditable absence. Expected inverter response absent under verified communications and power availability.

Metrics. Wh delivered per control joule; tracking efficiency; response time; thermal cycling.

Evidence: C0 R17 R31

ENG-06

Battery state-of-health and thermal-runaway guard engine

Near-term pilot**Addition.**

Role. Use continuous state relations, lineage of cells/modules and next-event guards for degradation, imbalance, venting and thermal risk.

Guard. Predicted temperature, voltage, pressure, impedance or imbalance crosses a certified threshold.

Auditable absence. Expected sensor/contact response absent; never interpret missing telemetry as safe.

Metrics. Prediction horizon; false/late trips; cycle life; pack availability; compute energy.

Evidence: C0

ENG-07**EV charging and fleet energy routing**

Near-term pilot

Addition.

Role. Use vehicle/charger/grid compatibility, local feeder support and next-departure/constraint events with lineage-aware energy accounting.

Guard. Charge/discharge transition admitted under connector, battery, departure, feeder and tariff constraints.

Auditable absence. Expected vehicle arrival, charger acknowledgment or departure readiness absent.

Metrics. Peak demand; missed departures; energy cost; battery wear; charger utilization.

Evidence: C0 R15

ENG-08**Gas, oil, hydrogen and CO2 pipeline leak/rupture event twin**

Near-term research pilot

Addition/upgrade to CPM and response coordination.

Role. Express leak/rupture, negative-pressure-wave and valve-response events with support, compatibility, uncertainty and auditable negative events.

Guard. Hydraulic/mass-balance or wave relation crosses a leak guard and is compatible with operating transients.

Auditable absence. Expected valve closure, pressure recovery or sensor report absent within a verified interval.

Metrics. Leak size/time sensitivity; location error; false alarms; isolated volume; response time.

Evidence: C0 R19 R20

ENG-09**Water distribution leak and service-loss event system**

Near-term municipal pilot

Addition.

Role. Combine hydraulic relations, local geographic support, meter/sensor compatibility and 311 reports; explicitly model service absence.

Guard. Pressure/flow imbalance or service complaint crosses a localized guard.

Auditable absence. Expected pressure, meter heartbeat or repair completion absent under coverage.

Metrics. Water loss; detection time; false alarms; repair time; equity by neighborhood.

Evidence: C0 R23 R24

ENG-10**Industrial combustion, boilers and process heat next-event optimization**

Research pilot

Addition.

Role. Layer a query-first event twin over validated thermodynamic/control models to predict limit crossings, fouling and efficiency transitions.

Guard. Efficiency, emissions, temperature or pressure relation crosses a guard under a candidate operating branch.

Auditable absence. Expected actuator or sensor response absent.

Metrics. Fuel per output; emissions; downtime; false interventions; control energy.

Evidence: C0

ENG-11**Fossil-fuel reduction through event-native operations**

Portfolio hypothesis

Cross-sector addition; not a substitute for clean-energy policy or physical transition.

Role. Apply event-native sensing, direct queries, provenance and compatibility to reduce avoidable compute, leaks, curtailment, idling and overproduction.

Guard. Avoided-energy or avoided-emission action occurs only when a verified event justifies it.

Auditable absence. Missing data cannot be counted as avoided use or emissions.

Metrics. Net J saved; kgCO₂e avoided; rebound; capital cost; verification confidence.

Evidence: C0 R15 R18 R19

ENG-12**Data-center and AI energy workload router**

Research prototype

Addition.

Role. Use local workload support, compatibility, event thresholds and hardware-specific energy models to choose electronic, analog or photonic paths.

Guard. Route a task when accuracy, latency, energy, precision and hardware calibration predicates are satisfied.

Auditable absence. Expected accelerator result or calibration signal absent; fall back.

Metrics. J/query; throughput; accuracy; data movement; fallback rate.

Evidence: C0 R32

ENG-13**Building HVAC and indoor-environment event twin**

Near-term pilot

Addition.

Role. Use local zone supports, occupancy/privacy compatibility and next-comfort/air-quality/fault events; store exceptions and novelty rather than every derived view.

Guard. Comfort, CO₂, humidity, equipment or demand guard crossing.

Auditable absence. Expected occupancy or sensor signal absent only with coverage/consent known.

Metrics. kWh; comfort; indoor air quality; false occupancy; maintenance.

Evidence: C0 R21 R24

ENG-14**Wind turbine and rotating-equipment next-fault engine**

Near-term pilot

Addition.

Role. Represent phase, rotational state, load and component lineage; solve bearing, imbalance and fatigue guard events.

Guard. Vibration/temperature/load relation crosses a calibrated fault guard.

Auditable absence. Expected phase marker or sensor heartbeat absent.

Metrics. Lead time; false alarms; downtime; maintenance cost; energy availability.

Evidence: C0

ENG-15

District heating, cooling and thermal-network routing

Research pilot

Addition.

Role. Use continuous transport relations, local branch support and compatibility to predict demand/temperature events and route heat.

Guard. Supply/return temperature, pressure or storage state crosses an operational guard.

Auditable absence. Expected demand or telemetry absent under verified coverage.

Metrics. Fuel/electricity; losses; comfort; peak reduction; event accuracy.

Evidence: C0

ENG-16

Carbon and energy provenance ledger

Implementable information-system pilot

Addition.

Role. Use lineage, versioned factors, compatibility and explicit missingness to make every claim reconstructable and branch-aware.

Guard. A carbon/energy claim is emitted only when source data, factor version, scope and ownership predicates are complete.

Auditable absence. Missing activity or factor data remains a declared gap, not a zero.

Metrics. Coverage; audit time; recalculation reproducibility; uncertainty; double-count prevention.

Evidence: C0 R28 R29

ENG-17

Renewable curtailment and storage counterfactual planner

Near-term planning pilot

Addition.

Role. Use isolated branches over a shared state to ask which intervention prevents the next constraint while preserving provenance.

Guard. Branch-specific constraint or reliability event under a candidate storage, network or demand-response action.

Auditable absence. Expected asset availability absent in a branch is explicit.

Metrics. Curtailment avoided; reliability; cost; model runtime; branch explainability.

Evidence: C0 R15 R16

ENG-18

Environmental sensing with event/absence semantics

Near-term pilot

Addition.

Role. Use local support, sensor provenance, coverage contracts and threshold events; distinguish non-detection from sensor silence.

Guard. Pollutant or ecological indicator crosses a calibrated guard under valid sensor support.

Auditable absence. Non-detection is emitted only when detection capability and sampling coverage are certified.

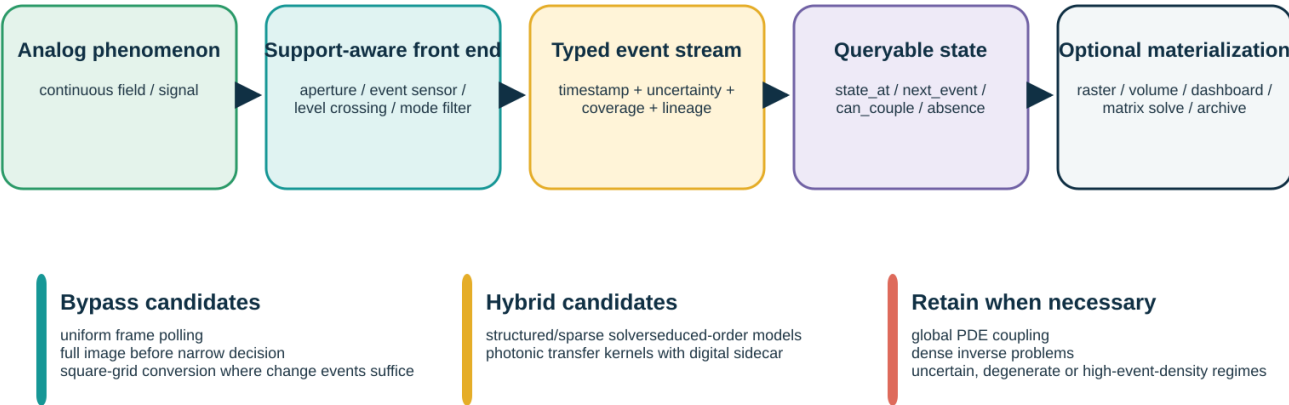
Metrics. Detection delay; false absence; sensor uptime; spatial coverage; energy.

Evidence: C0 R24

Digital, compute & data pipelines

Digital pipeline: from analog change to direct event semantics

Reduce conversion and materialization where possible; keep dense numerical methods when they remain the right solver



The target is less unnecessary materialization - not the elimination of linear algebra, ADCs or graphics.

Application overview: Digital, compute & data pipelines

DIG-01

Equation-world simulation kernel

Research prototype

Partial replacement of the authoritative simulation layer in bounded domains.

Role. Implement direct state_at, next_event, support, coupling, routing and lineage queries for a bounded relation family; render only on demand.

Guard. Earliest admitted relation crossing with certified root/interval.

Auditable absence. Expected event absent only with complete relation/support coverage.

Metrics. Query cost; event ordering; memory/novelty; correctness; expression growth.

Evidence: C0

DIG-02

Raymarching/rasterization bypass for state and collision

Research prototype

Partial replacement of state/collision pipeline; rendering remains optional.

Role. Move authority to relations/events and use ray/raster methods only as materialized views; collision and causality use direct queries.

Guard. Visibility, contact or transition query solved from relations rather than image sampling.

Auditable absence. No pixel or sample is never evidence that an entity is absent from state.

Metrics. State query time; collision correctness; render independence; memory; view consistency.

Evidence: C0 R33

DIG-03**Dense linear-algebra avoidance planner**

Research architecture

Addition/partial workload replacement, not elimination of linear algebra.

Role. Use support pruning, typed relations, sparse/structured solvers and task routing; use dense linear algebra only where it is the correct local kernel.

Guard. Choose the smallest valid solver/kernel whose support and precision satisfy the query.

Auditable absence. A solver result is unavailable when support or conditioning is insufficient.

Metrics. FLOPs avoided; data movement; J/query; accuracy; fallback rate.

Evidence: C0 R32

DIG-04**Event-driven analog-to-digital conversion layer**

Near-term hardware research

Addition/partial replacement in sparse sensing.

Role. Use level-crossing, change-triggered or feature-triggered conversion for sparse signals, with conventional sampling as fallback and calibration reference.

Guard. Analog signal crosses a calibrated level or derivative guard.

Auditable absence. No event means no threshold crossing only if sensor health, coverage and threshold configuration are known.

Metrics. Samples/signal; energy; reconstruction error; latency; missed crossings.

Evidence: C0 R31

DIG-05**Event-camera perception substrate**

Implementable research pilot

Addition/partial replacement for motion-centric tasks.

Role. Use asynchronous change events as input support, then compatibility and lineage to form object/interaction events; images remain optional.

Guard. Local contrast event joins a tracked entity/trajectory and crosses a task guard.

Auditable absence. No events imply no local contrast changes only within sensor support; not object absence.

Metrics. Latency; bandwidth; events/J; tracking accuracy; low-motion failure.

Evidence: C0 R30

DIG-06**Photonic/mixed-signal query accelerator**

Frontier prototype

Addition.

Role. Route stable, high-reuse linear transforms to photonic or analog kernels and keep compatibility, uncertainty, nonlinear decisions and lineage digital.

Guard. Hardware path admitted when precision, calibration, batch reuse and I/O energy satisfy a policy.

Auditable absence. Missing calibration/reference event forces digital fallback.

Metrics. End-to-end J/query; throughput; accuracy; I/O share; recalibration.

Evidence: C0 R32

DIG-07

Novelty-proportional state storage

Near-term software pilot**Partial replacement of snapshot-heavy history for closed dynamics.**

Role. Store seed/grammar, exogenous events, lineage, branch policy and checkpoints; materialize views on demand.

Guard. An exogenous intervention or irreversible observation is appended as novelty.

Auditable absence. Expected events can expire into absence events; missing coverage remains unknown.

Metrics. Storage growth; rebuild time; auditability; branch cost; view freshness.

Evidence: C0 R28

DIG-08

Auditable absence and negative-event ledger

Implementable design pattern**Addition to any event system.**

Role. Add ExpectedEvent and AbsenceEvent objects requiring obligation, observation support, deadline, coverage and reason codes.

Guard. At deadline, emit absence only if expectation exists, observation support was valid and no compatible event occurred.

Auditable absence. The pattern itself formalizes auditable absence.

Metrics. False absence; unknown-vs-absent classification; overdue time; coverage quality.

Evidence: C0 R6 R7 R24 R26

DIG-09

Data-pipeline state and lineage engine

Implementable integration**Addition/partial replacement of dashboard-centric orchestration.**

Role. Represent datasets, transformations, models, approvals and deployments as typed events with compatibility and provenance.

Guard. Transformation or release admitted only when inputs, schema, tests, authority and policy are compatible.

Auditable absence. Expected artifact, test or approval absent after a due interval.

Metrics. Reproducibility; failure diagnosis; stale-data incidents; audit time.

Evidence: C0 R28 R29

DIG-10

Cybersecurity trust-zone and policy event system

Near-term architecture pilot**Addition.**

Role. Use sheet/sector separation, versioned compatibility predicates, provenance and event guards; co-location never implies access.

Guard. Access or data join admitted only under identity, device, context, policy version and time predicates.

Auditable absence. Expected attestation or heartbeat absent triggers degraded trust, not automatic compromise.

Metrics. Unauthorized joins; policy explainability; response time; false blocks.

Evidence: C0 R28 R35

DIG-11

Distributed-system reconstruction and branch debugging

Near-term software pilot

Addition.

Role. Use explicit event lineage, branch context, compatibility and invariant reports to reconstruct what happened and why.

Guard. State transition creates before/after hashes, predicate version, branch and invariant report.

Auditable absence. Expected acknowledgment absent can be distinguished from delayed, dropped or never-sent.

Metrics. Mean time to resolution; replay fidelity; log volume; invariant violations.

Evidence: C0 R28

DIG-12

AI memory and retrieval with provenance and support

Near-term pilot

Addition.

Role. Use local semantic support, compatibility, provenance and event-based updates; render summaries only from admitted evidence.

Guard. Evidence enters a response only when relevance, authority, time, permission and provenance predicates pass.

Auditable absence. Expected evidence absent is reported as a gap, not fabricated.

Metrics. Citation accuracy; retrieval precision; stale-source rate; permission violations.

Evidence: C0 R28 R29

DIG-13

Model monitoring and deployment event graph

Implementable integration

Addition.

Role. Track data, model, evaluation, approval, deployment and drift as a provenance graph with compatibility and rollback branches.

Guard. Deployment admitted when dataset, model, evaluation, policy and environment are compatible.

Auditable absence. Expected monitoring metric absent triggers a hold, not assumed normality.

Metrics. Rollback time; drift detection; reproducibility; audit completeness.

Evidence: C0 R28 R29

DIG-14

Network and telecom event-native monitoring

Near-term pilot

Addition.

Role. Use local flow supports, compatibility and threshold events; retain raw data around exceptions and provenance.

Guard. Latency, loss, congestion, security or service relation crosses a guard.

Auditable absence. Expected heartbeat or route acknowledgment absent under verified coverage.

Metrics. Data reduction; detection time; false alarms; root-cause time; energy.

Evidence: C0

DIG-15

AR/VR and simulation foveated query substrate

Research prototype

Addition/partial state-pipeline replacement.

Role. Use gaze/agent-local spherical support for relevance and direct state/event queries; render only admitted views and preserve nonvisual state independently.

Guard. View or interaction event within gaze, reach, causal or semantic support.

Auditable absence. No rendered object is not evidence of world absence.

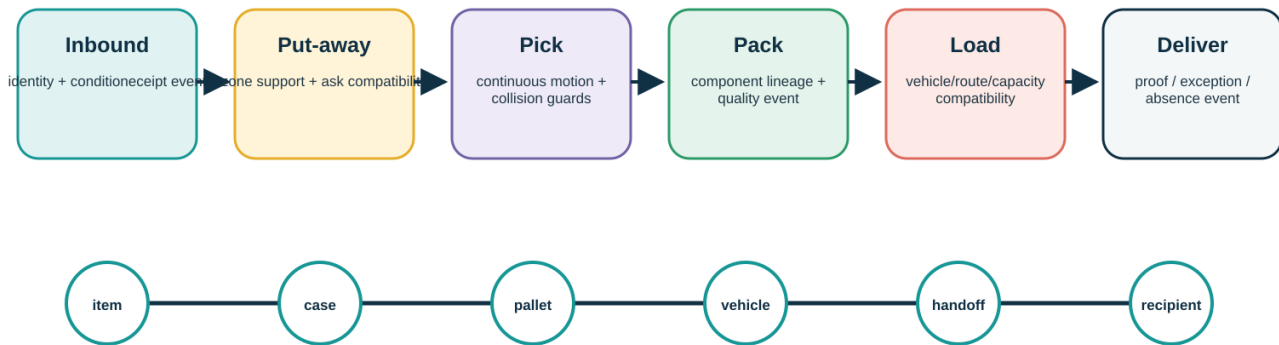
Metrics. Frame cost; motion-to-photon latency; interaction correctness; energy.

Evidence: C0

Warehousing, logistics & supply chains

Warehouse and delivery optimization as event lineage

Items, containers, tasks, vehicles and custody changes are explicit relations - not inferred from stale coordinates



Lineage records splits, merges, substitutions and custody - while ExpectedEvent/AbsenceEvent distinguishes late, missing and unobserved.

Application overview: Warehousing, logistics & supply chains

LOG-01

Warehouse event twin and inventory truth

Implementable pilot

Addition beneath WMS/WES.

Role. Use EPCIS-like visibility events, lineage and auditable absence to reconstruct item, tote, pallet, location and task state.

Guard. Receive, move, pick, pack, load, count and exception events update lineage.

Auditable absence. Expected scan or movement absent under verified reader/task coverage.

Metrics. Inventory accuracy; search time; write-offs; reconciliation labor; event completeness.

Evidence: C0 R26 R27

LOG-02

AMR and forklift continuous routing

Near-term pilot

Addition.

Role. Use continuous trajectories, local supports, compatibility and next-conflict events across vehicles, humans and temporary zones.

Guard. Earliest predicted conflict, blockage or task-window crossing.

Auditable absence. Tracking absence blocks high-speed motion; no return is not free space.

Metrics. Throughput; near misses; stops; travel distance; energy/order.

Evidence: C0 R33 R34

LOG-03**Pick-path and wave planning as event scheduling**

Near-term software pilot

Addition.

Role. Schedule next admissible pick/pack/ship events using local resource support, due-time guards and branch isolation.

Guard. Task admitted when inventory, worker/robot, container, dock and promise constraints match.

Auditable absence. Expected inventory or resource event absent.

Metrics. Orders/hour; lateness; walking/drive distance; replans; fairness.

Evidence: C0

LOG-04**Cold-chain and condition lineage**

Implementable pilot

Addition.

Role. Bind sensor observations, packaging, custody and product lineage; trigger excursion and missing-coverage events.

Guard. Temperature/humidity/time relation exits product support during a custody interval.

Auditable absence. Expected logger data absent under a known custody and power window.

Metrics. Excursion detection; spoilage; false holds; audit time.

Evidence: C0 R24 R26 R27

LOG-05**Last-mile delivery event and promise engine**

Near-term pilot

Addition.

Role. Use continuous vehicle/package state, local traffic support, compatibility and delivery-promise guards with explicit non-delivery reasons.

Guard. Delivery becomes admissible when parcel, recipient/location, vehicle, access and time conditions match.

Auditable absence. Expected arrival, handoff or proof event absent with verified route/coverage.

Metrics. On-time rate; miles/order; failed deliveries; customer contacts; emissions.

Evidence: C0 R26 R27

LOG-06**Port, terminal and yard coordination**

Research/operations pilot

Addition.

Role. Create compatibility-gated events and lineage across berth, yard, crane, customs and onward transport.

Guard. Move admitted when asset, slot, clearance, equipment and safety states match.

Auditable absence. Expected clearance, equipment or handoff event absent.

Metrics. Dwell time; crane productivity; truck queues; misroutes; energy.

Evidence: C0 R26 R27

LOG-07

Food provenance and recall event graph

Implementable integration

Addition.

Role. Use critical tracking events, split/merge lineage, condition data and compatibility to localize affected product.

Guard. Transformation, aggregation, shipment, receipt and recall events preserve parent/child lineage.

Auditable absence. Expected custody or transformation record absent.

Metrics. Recall scope; trace time; waste; coverage; false inclusion/exclusion.

Evidence: C0 R26 R27

LOG-08

Medical-device and medicine supply chain

Implementable pilot

Addition.

Role. Combine GS1 traceability with hospital compatibility and patient/procedure context.

Guard. Device/drug use admitted when product, lot, expiry, storage, order and patient/procedure predicates pass.

Auditable absence. Expected receipt, scan, temperature or recall acknowledgment absent.

Metrics. Expired/incorrect use; recall time; stockouts; auditability.

Evidence: C0 R6 R26 R27

LOG-09

Returns, repair and circular-economy lineage

Near-term pilot

Addition.

Role. Preserve component lineage, condition events, repair branches and material provenance to route reuse, refurbishment or recycling.

Guard. Disposition admitted when identity, condition, warranty, safety and material predicates pass.

Auditable absence. Expected inspection or test result absent.

Metrics. Recovered value; waste avoided; turnaround; traceability.

Evidence: C0 R26 R28

LOG-10

Supply-chain risk and counterfactual branch planner

Research pilot

Addition.

Role. Use isolated branches sharing a base lineage to ask next disruption, substitute compatibility and downstream impact.

Guard. Supplier, lane, inventory or regulatory relation crosses a disruption guard.

Auditable absence. Expected shipment or supplier confirmation absent under a known reporting contract.

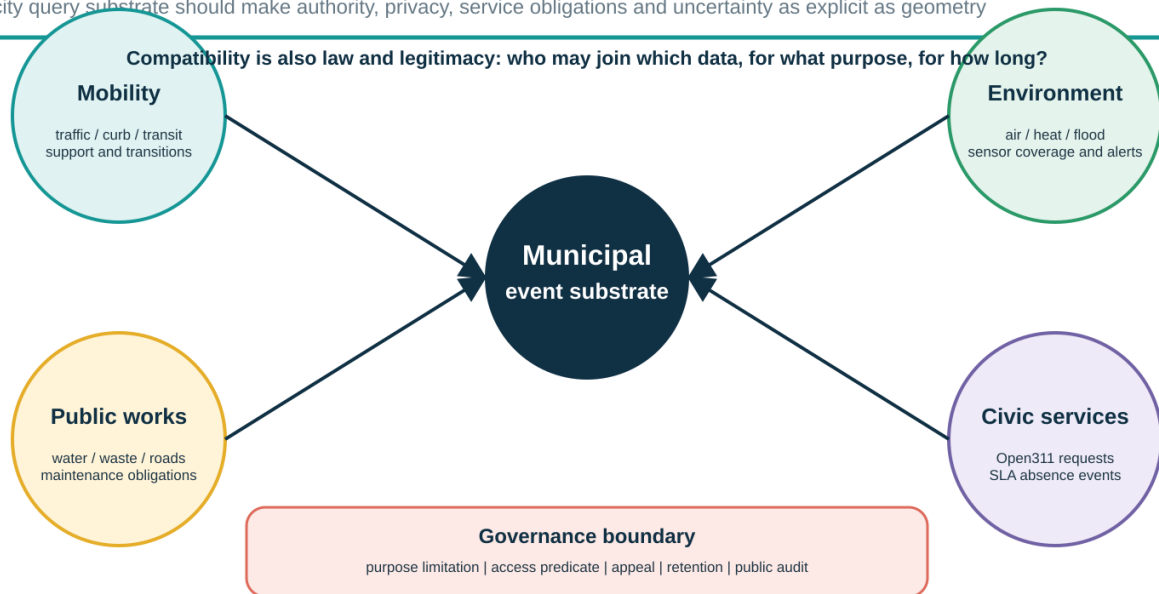
Metrics. Time to detect; recovery time; substitution success; scenario explainability.

Evidence: C0 R26 R27

Smart cities & municipal governance

Rights-preserving smart city and municipal governance

A city query substrate should make authority, privacy, service obligations and uncertainty as explicit as geometry



Application overview: Smart cities & municipal governance

CIV-01 Municipal service obligation and 311 event graph

Implementable pilot

Addition beneath CRM/asset systems.

Role. Link requests, assets, departments, contractors, SLAs, inspections and resident communications through events and auditable absences.

Guard. Request, dispatch, work, inspection and closure are typed events with reason and evidence.

Auditable absence. Expected response, visit or resolution absent after its service window.

Metrics. Resolution time; reopen rate; equity; evidence completeness; resident trust.

Evidence: C0 R21 R23 R25

CIV-02 Transit arrival and missed-service ledger

Near-term pilot

Addition.

Role. Represent scheduled/expected service, vehicle lineage, stop support and observed arrivals; emit absence only with coverage.

Guard. Arrival/departure crosses stop/time support under a vehicle/run identity.

Auditable absence. Scheduled trip absent when AVL and operational coverage are verified; otherwise unknown/data gap.

Metrics. On-time performance; ghost-trip classification; passenger delay; data coverage.

Evidence: C0 R21 R22 R24

CIV-03

Traffic, curb and intersection event control

Research pilot

Addition.

Role. Use local spatial supports, vehicle/pedestrian compatibility and next-conflict events; preserve policy versions and human override.

Guard. Conflict, queue, pedestrian call, emergency preemption or curb-occupancy guard crossing.

Auditable absence. Expected detector or phase acknowledgment absent.

Metrics. Delay; safety surrogate; emissions; false calls; accessibility.

Evidence: C0 R21 R24 R33

CIV-04

Water, sewer, waste and street-asset lifecycle

Near-term integration

Addition.

Role. Give each asset lineage, condition events, local support and maintenance guards; connect physical and budget records.

Guard. Inspection, fault, maintenance, replacement and service events update asset state.

Auditable absence. Expected inspection, collection or repair absent under a service obligation.

Metrics. Failure rate; backlog; lifecycle cost; service equity; auditability.

Evidence: C0 R21 R22 R24 R28

CIV-05

Emergency response common event substrate

Research/operations pilot

Addition.

Role. Use a common incident lineage, local support, compatibility and event guards while preserving agency-specific sectors and permissions.

Guard. Incident, dispatch, arrival, hazard, capacity and handoff events are admitted under role and jurisdiction predicates.

Auditable absence. Expected unit acknowledgment or arrival absent within an explicit response window.

Metrics. Response time; duplicate dispatch; handoff quality; privacy incidents.

Evidence: C0 R21 R24 R35

CIV-06

Permitting, licensing and inspection event pipeline

Implementable pilot

Addition/partial replacement of opaque workflow tracking.

Role. Model application, evidence, review, inspection, decision, appeal and policy version as events with explicit compatibility and reason codes.

Guard. Decision transition admitted only when required evidence, authority, policy version and procedural rights are satisfied.

Auditable absence. Expected review or applicant response absent after due interval, with waiver/extension states.

Metrics. Cycle time; rework; appeals; missing documents; transparency.

Evidence: C0 R28

CIV-07

Budget, procurement and contract lineage

Implementable information-system pilot**Addition.**

Role. Create lineage from policy goal to appropriation, tender, bid, award, change order, invoice, delivery and outcome.

Guard. Financial or contractual transition carries authority, policy, evidence and before/after state.

Auditable absence. Expected deliverable, report or payment evidence absent.

Metrics. Audit time; change-order visibility; duplicate spend; outcome linkage.

Evidence: C0 R28 R29

CIV-08

Policy simulation with isolated counterfactual branches

Near-term planning pilot**Addition.**

Role. Use branch-isolated policy worlds sharing base data; compare next events, service KPIs and distributional impacts without overwriting live state.

Guard. Policy-specific service, cost, risk or equity guard crossing.

Auditable absence. Missing evidence is shown as uncertainty, not zero impact.

Metrics. Scenario reproducibility; runtime; KPI coverage; distributional impact; explainability.

Evidence: C0 R22 R25

CIV-09

Rights-preserving municipal data spaces

Implementable architecture pilot**Addition and governance foundation.**

Role. Use relational exclusion, purpose/time/role compatibility, provenance and public reason codes; co-location in a platform never implies join permission.

Guard. Data join admitted only for declared purpose, authority, minimization, retention and consent/legal basis.

Auditable absence. Expected deletion, review or access-log event absent triggers governance action.

Metrics. Unauthorized joins; purpose violations; access audit; resident redress.

Evidence: C0 R21 R35

CIV-10

Community well-being and service-outcome atlas

Near-term analytics pilot**Addition.**

Role. Map technology events through infrastructure services to community benefits, with lineage and neighborhood support rather than one global score.

Guard. Service outcome or disparity crosses a declared community guard.

Auditable absence. Expected service result absent, with data coverage and denominator explicit.

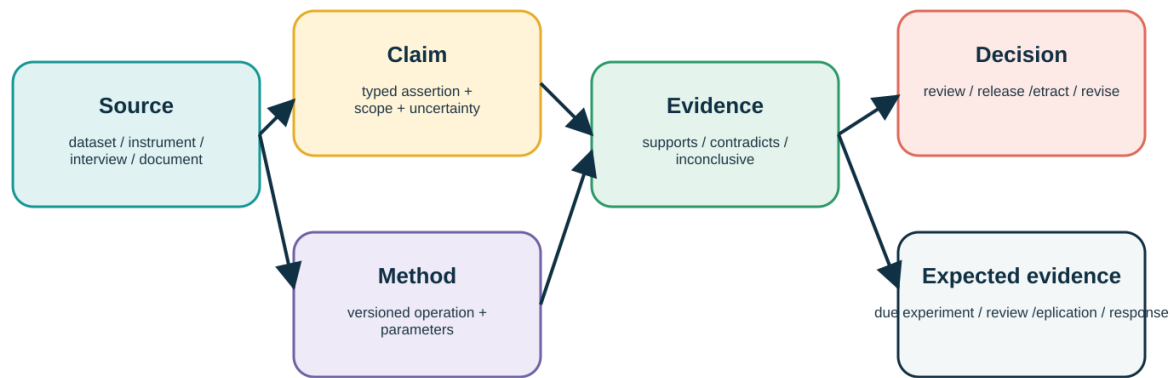
Metrics. Service access; quality of life; equity; resilience; data completeness.

Evidence: C0 R22 R25

Research & documentation management

Research and documentation management as executable provenance

Documents remain human views; claims, evidence, transformations, decisions and gaps form the authoritative graph



Materialized views
paper, PDF, dashboard, registry entry and literature review are generated from the provenance graph; staleness and omissions are explicit.

Application overview: Research & documentation management

RES-01

Claim-evidence-provenance graph

Implementable pilot

Addition beneath document systems.

Role. Represent claims, evidence, analyses, authorship, review and revision as a provenance graph; documents become materialized views.

Guard. Claim created, supported, challenged, revised or withdrawn with source and method lineage.

Auditable absence. Expected evidence, review or replication absent is explicit.

Metrics. Traceability; review time; unsupported claims; reproducibility.

Evidence: C0 R28 R29

RES-02

Negative-result and non-observation registry

Implementable design pattern

Addition.

Role. Use ExpectedEvent/AbsenceEvent with protocol, detection support, power, deadline and reason to record meaningful non-observation.

Guard. A pre-registered expected observation expires under adequate detection support without a compatible event.

Auditable absence. This is the core absence use case.

Metrics. Publication bias; repeat failures; power/coverage quality; reuse.

Evidence: C0 R29

RES-03

Electronic lab notebook as event-sourced experiment

Implementable pilot

Addition/partial replacement of page-centric ELN history.

Role. Record protocol, material, instrument, operator, environment, observation and transformation events with lineage.

Guard. Experiment transition emits before/after state, instrument calibration, sample lineage and confidence.

Auditable absence. Expected run, calibration or sample event absent.

Metrics. Reproducibility; audit time; missing metadata; onboarding.

Evidence: C0 R28 R29

RES-04

Living systematic review and evidence map

Near-term software pilot

Addition.

Role. Maintain study, population, intervention, outcome, bias and claim events; materialize current reviews by query.

Guard. Study or result changes the support for a claim under a versioned inclusion predicate.

Auditable absence. Expected registry result or follow-up publication absent is tracked as pending/unknown.

Metrics. Update latency; citation coverage; reproducibility; conflict visibility.

Evidence: C0 R28 R29

RES-05

Regulatory and quality-management evidence bundle

Implementable integration

Addition.

Role. Use event lineage and compatibility to generate auditable, version-specific evidence bundles.

Guard. Requirement, design, test, deviation, risk control and approval events link to a release.

Auditable absence. Expected verification or approval absent blocks release.

Metrics. Audit prep time; trace gaps; change impact; defect escape.

Evidence: C0 R28

RES-06

Software and computational-environment lineage

Implementable pilot

Addition.

Role. Treat environment and execution as lineage events; reconstruct exact branch and artifacts for each result.

Guard. Run event links code commit, dependencies, data, hardware, parameters and outputs.

Auditable absence. Expected artifact or checksum absent.

Metrics. Reproduction success; build time; provenance completeness.

Evidence: C0 R28 R29

RES-07

Research portfolio and funding lineage

Near-term information-system pilot

Addition.

Role. Build lineage from question and funding to experiments, assets, outputs, reuse and societal outcomes without collapsing evaluation into one score.

Guard. Funding, milestone, dataset, publication, reuse and outcome events preserve attribution.

Auditable absence. Expected deliverable or data release absent with documented reason.

Metrics. Duplication; reuse; time to evidence; administrative burden.

Evidence: C0 R28 R29

RES-08

Knowledge transfer and institutional memory substrate

Near-term pilot

Addition.

Role. Store decisions, assumptions, predicates, evidence and outcomes as lineage-aware events; documents and dashboards are generated views.

Guard. Decision or rule change records context, authority, evidence, alternatives and consequences.

Auditable absence. Expected review or retirement event absent.

Metrics. Onboarding time; repeated mistakes; decision traceability; stale rules.

Evidence: C0 R28

Replacement and addition matrix

A quick boundary map for high-profile systems and bottlenecks.

Current system / layer	Disposition	Why	Enabled addition	Evidence boundary
MRI magnet, gradients and RF chain	Retain	Physical signal generation and safety remain governed by MRI physics and regulation.	Query-first sequence control, adaptive support, direct quantitative parameter estimation and lineage-aware workflow.	Full replacement unsupported; narrow task-specific MR instruments are a frontier.
MRI image reconstruction	Partially replace for bounded tasks	Images remain valuable and clinically entrenched.	Direct parameter or event queries can precede or bypass full images for validated narrow questions.	Must compare diagnostic performance, uncertainty, scan time and safety.
ASML EUV projection tool	Retain	The event calculus does not create 13.5 nm light, mirrors, stages or resist chemistry.	Add event-native control, local hotspot solving, lineage and photonic metrology kernels.	Claims that slower chips make EUV unnecessary are rejected without system evidence.
Computational lithography/OPC/SMO	Augment and selectively replace	Current tools already use physics and optimization.	Use local support, typed guards, reduced-order relation families and branch provenance.	Must preserve full-chip and long-range effects.
EHR database and documents	Add a substrate beneath	FHIR resources and documents remain exchange and presentation standards.	Use a person lineage, event graph, consent/compatibility and auditable absence across life stages.	Never reduce the person to a diagnosis or one snapshot.
Hospital dashboards	Partially replace as authority	Dashboards are views and can be stale.	Derive them from event/lineage state and explicit expected actions.	Clinical judgment and operational governance remain human.
Uniform-sampling ADC	Partially replace for sparse signals	Uniform sampling is robust and general.	Use level-crossing/event conversion with reference sampling and fallback.	No event is meaningful only with verified coverage.
Raster/raymarch world store	Replace as authority in bounded worlds	Raster/ray methods remain useful for display.	Store/query relations, events and lineage; materialize images only when needed.	Hard geometry may still require conventional solvers.
Dense global linear algebra	Reduce, route and hybridize	Linear algebra remains essential.	Avoid unnecessary global tensors through support pruning, structured solvers and analog/photonic kernels.	End-to-end I/O, precision and calibration decide value.
SCADA/historian	Add event twin and provenance	Historians remain important evidence sources.	Compile continuous state, next-event guards, expected responses and lineage.	Do not discard raw evidence around incidents.
Pipeline CPM	Augment	CPM already models pressure, flow, temperature and equipment.	Add explicit compatibility, uncertainty, negative events and response lineage.	Must meet regulation and controlled validation.
PV monitoring and inverter control	Augment	Established grid-control standards remain authoritative.	Add local support, event-native fault detection, lineage and events/j metrics.	Low-level protections remain certified.
WMS/WES inventory snapshots	Add/partially replace state history	Operational systems remain transaction engines.	Use visibility events, lineage and expected scans to reconstruct physical truth.	Reader coverage and human exceptions must be modeled.
Municipal CRM/311	Add obligation and evidence layer	Existing channels remain citizen interfaces.	Link service request, asset, SLA, work, inspection, closure and absence.	Prevent closure gaming and digital exclusion.
City data lake	Add purpose-separated event spaces	Data platforms remain useful storage.	Use compatibility/purpose predicates, provenance, branch isolation and local community support.	Avoid surveillance and unauthorized joins.
Mental-health record	Augment	Clinical care cannot be automated by geometry.	Use patient-controlled longitudinal events, dimensional measures and consent.	No inference from silence; no autonomous diagnosis or treatment.
Research paper/PDF as sole record	Replace as sole authority	Documents remain human-readable views.	Maintain claim-evidence-provenance graph and generate documents from it.	Scholarly interpretation and debate remain human.

Current system / layer	Disposition	Why	Enabled addition	Evidence boundary
General-purpose optical computer	Do not claim	Photonic components can accelerate stable transforms.	Use narrow calibrated relation kernels with digital sidecar and fallback.	Zero heat, zero power and universal computation are rejected.

Metric dictionary

Candidate measures for task, event, energy, reliability, governance and externality evaluation.

Metric	Unit / definition	Interpretation
Theta_verified	verified events / second	Events satisfying support, compatibility, guard and confidence.
Theta_E	verified events / joule	Includes sensing/source, conversion, actuation, control, calibration and storage.
eta_route	verified / admitted	Correct routing among admitted candidates.
R_support	all candidates / supported candidates	Pruning due to local radial/angular/causal/semantic support.
R_compat	supported / solver-entering candidates	Pruning due to physical, policy, identity and mode compatibility.
L50	median event latency	Median time from physical or digital condition to verified event.
L99	99th percentile event latency	Tail latency including settling, network and control.
P_miss	missed-event probability	Probability of failing to emit a required event.
P_false	false-event probability	Probability of emitting an unjustified event.
P_abs_false	false-absence probability	Absence emitted despite an unobserved or late compatible event.
P_unknown	unknown/missing-coverage fraction	Cases that cannot be classified as present or absent.
coverage_support	time/space within certified observation support	Required for meaningful negative events.
lineage_completeness	reconstructable transitions / required transitions	Coverage of parent/child, split/merge and provenance edges.
replay_fidelity	reconstructed state matches reference	Correctness of seed+grammar+log reconstruction.
branch_growth	active branch nodes / scenario	Counterfactual growth and pruning burden.
expression_growth	relation complexity vs grammar depth	Algebraic closure and normalization burden.
solver_work	root/constraint operations per verified event	Core event-solver cost.
materialization_avoided	bytes or operations not materialized	State/view work avoided by direct queries.
calibration_burden	time and energy per calibration interval	May erase analog/photonic advantage.
drift	transfer error vs time/temperature	Stability of physical and learned models.
fallback_rate	queries sent to conventional reference	Measures boundedness and practical coverage.
diagnostic_task_perf	sensitivity, specificity, AUC or calibrated utility	Biomedical outcome at a declared task.
scan_time	minutes per completed diagnostic task	MRI/imaging operational metric.
dose_or_exposure	radiation, RF/SAR, optical or phototoxic exposure	Physical acquisition burden.
overlay_focus_dose	semiconductor process accuracy metrics	Lithography-specific performance.
wafers_per_hour	completed wafers / hour at quality	Manufacturing throughput.
Wh_per_control_J	delivered energy per control-system joule	Solar/grid controller efficiency.
curtailment_avoided	MWh or percent	Renewable utilization impact.
leak_detect_time	time from leak onset to verified alarm	Pipeline/water safety.
isolated_volume	material released before isolation	Pipeline consequence metric.

Metric	Unit / definition	Interpretation
inventory_accuracy	correct physical/system matches	Warehouse truth metric.
orders_per_hour	verified orders completed / hour	Warehouse throughput.
miles_per_delivery	distance / completed delivery	Logistics efficiency.
service_sla_met	municipal obligations met / due	Municipal service outcome.
equity_gap	difference across neighborhoods or groups	Must be contextual and privacy-preserving.
audit_reconstruction_time	time to reconstruct a decision or event	Governance/research value.
reproduction_success	independent reruns matching declared result	Research reproducibility.
kgCO2e_net	verified lifecycle greenhouse-gas change	Includes rebound and system boundary.

Auditable absence examples

Cross-domain examples of the obligation, coverage, deadline and caveat needed before recording a missing presence.

Domain	Expectation	Required support	Deadline condition	Emitted state	Caveat
Hospital follow-up	Follow-up visit or contact due	Care pathway, consent, contact channels valid	Due interval closes with no compatible contact/visit	Absent follow-up	Do not emit if patient opted out, transferred care or contact coverage failed.
MRI sequence	Physiology or calibration event required	Sensor attached, channel healthy, sequence window active	Window closes with no event	Missing required acquisition support	Result is inconclusive, not negative anatomy.
Warehouse	Expected pallet scan at zone exit	Reader coverage and task assignment valid	Pallet exits time window without scan	Unobserved movement or process exception	Distinguish reader failure from item absence.
Delivery	Expected handoff by promise window	Vehicle, parcel and route tracking valid	Window closes without proof or exception	Failed/late delivery	Recipient unavailability is a reason code, not silent failure.
Pipeline	Expected valve closure after rupture identification	Valve command path and feedback sensor healthy	Required time closes without closed-state event	Mitigation failure	Escalate; do not infer from pressure alone.
Solar	Expected generation under irradiance support	Irradiance, inverter availability and telemetry valid	Generation relation not observed	Underperformance/fault candidate	Night, curtailment and telemetry outage are separate sectors.
Transit	Scheduled trip arrival	AVL coverage and trip not cancelled	Stop window closes without arrival	Missed trip	Unknown if AVL is down or detour unreported.
Municipal service	Inspection or repair promised	Work order active, access available	SLA interval closes	Overdue service	Waiver, resident access failure and weather delay are typed reasons.
Research	Pre-registered observation expected	Protocol power and instrument detection support valid	Experiment completes without event	Supported null/non-observation	Instrument failure or underpowered design remains unknown.
Data pipeline	Expected artifact/test/ approval	Job and repository available	Deadline closes without artifact	Blocked release	Differentiate queued, failed, skipped and never required.

Evidence registry and bibliography

External references establish components, standards and baselines. They do not validate the application atlas as an integrated system.

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Use in atlas: Clinical role and soft-tissue imaging baseline.
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5. **R4 - Magnetic resonance fingerprinting.** Ma et al. (2013). <https://doi.org/10.1038/nature11971>
Use in atlas: Direct quantitative tissue-parameter estimation precedent.
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11. **R10 - WHO guideline: recommendations on digital interventions for health system strengthening.** World Health Organization (2019). <https://www.who.int/publications/i/item/9789241550505>
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12. **R11 - Research Domain Criteria (RDoC) Initiative.** NIMH (current). <https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc>
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13. **R12 - Computational lithography.** ASML (current). <https://www.asml.com/products/computational-lithography>
Use in atlas: Scanner, mask and process co-optimization baseline.
14. **R13 - EUV lithography systems.** ASML (current). <https://www.asml.com/products/euv-lithography-systems>
Use in atlas: EUV/High-NA physical projection system baseline.
15. **R14 - ASML and imec open joint High NA EUV Lithography Lab.** ASML and imec (2024). <https://www.asml.com/news/press-releases/2024/asml-imec-opening-high-na-euv-lithography-lab>
Use in atlas: Current High-NA ecosystem and process-development context.
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Final boundary: this report broadens the application program beyond traditional sector lines while preserving the source report's central discipline: use the implementable operator, keep physical law and human authority explicit, and demote totalizing claims until a matched experiment earns them.